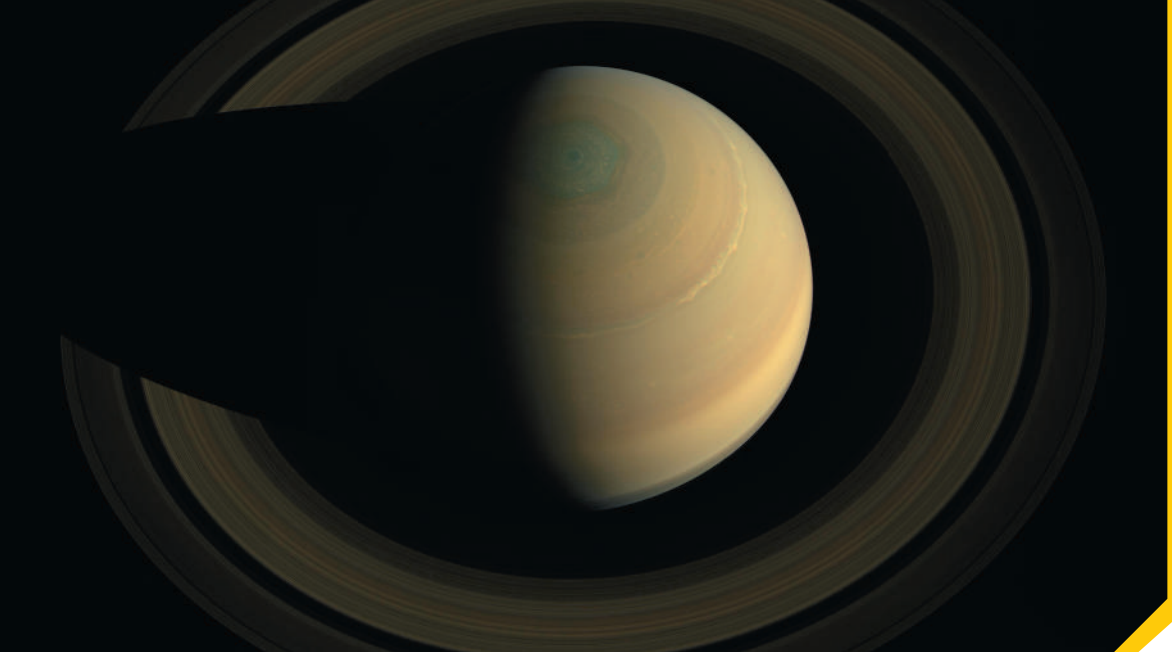




? The rings of Saturn are made of countless individual orbiting particles. Compared with a ring particle that orbits far from Saturn, does a ring particle close to Saturn orbit with (i) the same speed and greater acceleration; (ii) a faster speed and the same acceleration; (iii) a slower speed and the same acceleration; (iv) a faster speed and greater acceleration; or (v) none of these?

13 Gravitation

Some of the earliest investigations in physical science started with questions that people asked about the night sky. Why doesn't the moon fall to earth? Why do the planets move across the sky? Why doesn't the earth fly off into space rather than remaining in orbit around the sun? The study of gravitation provides the answers to these and many related questions.

As we remarked in Chapter 5, gravitation is one of the four classes of interactions found in nature, and it was the earliest of the four to be studied extensively. Newton discovered in the 17th century that the same interaction that makes an apple fall out of a tree also keeps the planets in their orbits around the sun. This was the beginning of *celestial mechanics*, the study of the dynamics of objects in space. Today, our knowledge of celestial mechanics allows us to determine how to put a satellite into any desired orbit around the earth or to choose just the right trajectory to send a spacecraft to another planet.

In this chapter you'll learn the basic law that governs gravitational interactions. This law is *universal*: Gravity acts in the same fundamental way between the earth and your body, between the sun and a planet, and between a planet and one of its moons. We'll apply the law of gravitation to phenomena such as the variation of weight with altitude, the orbits of satellites around the earth, and the orbits of planets around the sun.

13.1 NEWTON'S LAW OF GRAVITATION

The gravitational attraction that's most familiar to you is your *weight*, the force that attracts you toward the earth. By studying the motions of the moon and planets, Newton discovered a fundamental **law of gravitation** that describes the gravitational attraction between *any* two objects. Newton published this law in 1687 along with his three laws of motion. In modern language, it says

NEWTON'S LAW OF GRAVITATION Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 13.1 How to calculate the gravitational forces that any two objects exert on each other.
- 13.2 How to relate the weight of an object to the general expression for gravitational force.
- 13.3 How to use and interpret the generalized expression for gravitational potential energy.
- 13.4 How to calculate the speed, orbital period, and total mechanical energy of a satellite in a circular orbit.
- 13.5 How to apply and interpret the three laws that describe the motion of planets.
- 13.6 Why the gravitational force exerted by a spherically symmetric planet is the same as if all of the planet's mass were concentrated at its center.
- 13.7 How the earth's rotation affects the apparent weight of an object at different latitudes.
- 13.8 What black holes are, how to calculate their properties, and how astronomers discover them.

You'll need to review...

- 3.3 Projectile motion.
- 4.4 Weight.
- 5.2 Apparent weightlessness.
- 5.4 Force and acceleration in uniform circular motion.
- 5.5 The fundamental forces of nature.
- 7.1 Gravitational potential energy and total mechanical energy conservation.
- 7.4 Force and potential energy.

Figure 13.1 The gravitational forces that two particles of masses m_1 and m_2 exert on each other.

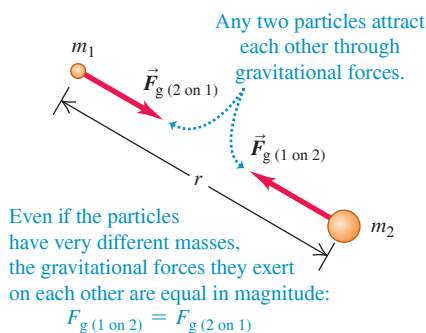


Figure 13.1 depicts this law, which we can express as an equation:

Newton's **law of gravitation**:
Magnitude of attractive gravitational force between any two particles

$$F_g = \frac{Gm_1m_2}{r^2}$$

Gravitational constant (same for any two particles)
Masses of particles
Distance between particles

(13.1)

The **gravitational constant** G in Eq. (13.1) is a fundamental physical constant that has the same value for *any* two particles. We'll see shortly what the value of G is and how this value is measured.

Equation (13.1) tells us that the gravitational force between two particles decreases with increasing distance r : If the distance is doubled, the force is only one-fourth as great, and so on. Although many of the stars in the night sky are far more massive than the sun, they are so far away that their gravitational force on the earth is negligibly small.

CAUTION Don't confuse g and G The symbols g and G are similar, but they represent two very different gravitational quantities. Lowercase g is the acceleration due to gravity, which relates the weight w of an object to its mass m : $w = mg$. The value of g is different at different locations on the earth's surface and on the surfaces of other planets. By contrast, capital G relates the gravitational force between any two objects to their masses and the distance between them. We call G a *universal* constant because it has the same value for any two objects, no matter where in space they are located. We'll soon see how the values of g and G are related. ■

Gravitational forces always act along the line joining the two particles and form an action–reaction pair. Even when the masses of the particles are different, the two interaction forces have equal magnitude (Fig. 13.1). The attractive force that your body exerts on the earth has the same magnitude as the force that the earth exerts on you. When you fall from a diving board into a swimming pool, the entire earth rises up to meet you! (You don't notice this because the earth's mass is greater than yours by a factor of about 10^{23} . Hence the earth's acceleration is only 10^{-23} as great as yours.)

Gravitation and Spherically Symmetric Objects

We have stated the law of gravitation in terms of the interaction between two *particles*. It turns out that the gravitational interaction of any two objects that have *spherically symmetric* mass distributions (such as solid spheres or spherical shells) is the same as though we concentrated all the mass of each at its center, as in **Fig. 13.2**. Thus, if we model the earth as a spherically symmetric object with mass m_E , the force it exerts on a particle or on a spherically symmetric object with mass m , at a distance r between centers, is

$$F_g = \frac{Gm_E m}{r^2} \quad (13.2)$$

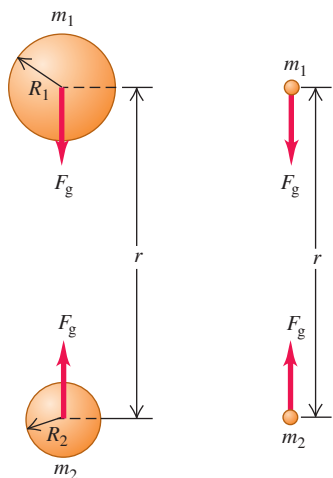
provided that the object lies outside the earth. A force of the same magnitude is exerted *on* the earth by the object. (We'll prove these statements in Section 13.6.)

At points *inside* the earth the situation is different. If we could drill a hole to the center of the earth and measure the gravitational force on an object at various depths, we would find that toward the center of the earth the force *decreases*, rather than increasing as $1/r^2$. As the object enters the interior of the earth (or other spherical object), some of the earth's mass is on the side of the object opposite from the center and pulls in the opposite direction. Exactly at the center, the earth's gravitational force on the object is zero.

Spherically symmetric objects are an important case because moons, planets, and stars all tend to be spherical. Since all particles in an object gravitationally attract each other, the particles tend to move to minimize the distance between them. As a result, the object naturally tends to assume a spherical shape, just as a lump of clay forms into a sphere if

Figure 13.2 The gravitational effect *outside* any spherically symmetric mass distribution is the same as though all of the mass were concentrated at its center.

- (a) The gravitational force between two spherically symmetric masses m_1 and m_2 ...
(b) ... is the same as if we concentrated all the mass of each sphere at the sphere's center.



you squeeze it with equal forces on all sides. This effect is greatly reduced in celestial objects of low mass, since the gravitational attraction is less, and these objects tend *not* to be spherical (**Fig. 13.3**).

Determining the Value of G

To determine the value of the gravitational constant G , we have to *measure* the gravitational force between two objects of known masses m_1 and m_2 at a known distance r . The force is extremely small for objects that are small enough to be brought into the laboratory, but it can be measured with an instrument called a *torsion balance*, which Sir Henry Cavendish used in 1798 to determine G .

Figure 13.4 shows a modern version of the Cavendish torsion balance. A light, rigid rod shaped like an inverted T is supported by a very thin, vertical quartz fiber. Two small spheres, each of mass m_1 , are mounted at the ends of the horizontal arms of the T. When we bring two large spheres, each of mass m_2 , to the positions shown, the attractive gravitational forces twist the T through a small angle. To measure this angle, we shine a beam of light on a mirror fastened to the T. The reflected beam strikes a scale, and as the T twists, the reflected beam moves along the scale.

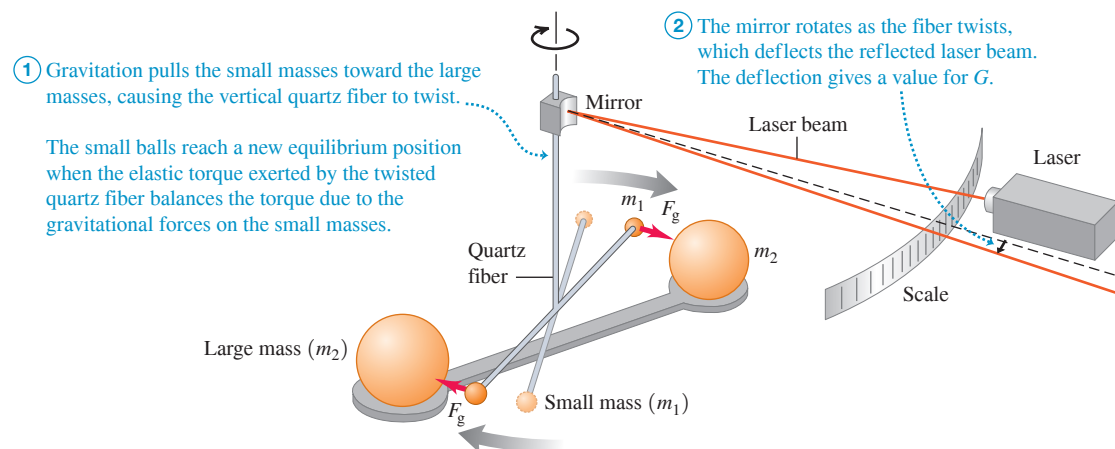
After calibrating the Cavendish balance, we can measure gravitational forces and thus determine G . The accepted value as of this writing (2018) is

$$G = 6.67408(31) \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$$

To three significant figures, $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$. Because $1 \text{ N} = 1 \text{ kg} \cdot \text{m}/\text{s}^2$, the units of G can also be expressed as $\text{m}^3/(\text{kg} \cdot \text{s}^2)$.

Gravitational forces combine vectorially. If each of two masses exerts a force on a third, the *total* force on the third mass is the vector sum of the individual forces of the first two. Example 13.3 makes use of this property, which is often called *superposition of forces* (see Section 4.1).

Figure 13.4 The principle of the Cavendish balance, used for determining the value of G . The angle of deflection has been exaggerated here for clarity.



EXAMPLE 13.1 Calculating gravitational force

The mass m_1 of one of the small spheres of a Cavendish balance is 0.0100 kg, the mass m_2 of the nearest large sphere is 0.500 kg, and the center-to-center distance between them is 0.0500 m. Find the gravitational force F_g on each sphere due to the other.

IDENTIFY, SET UP, and EXECUTE Because the spheres are spherically symmetric, we can calculate F_g by treating them as *particles* separated by 0.0500 m, as in Fig. 13.2. Each sphere experiences the same

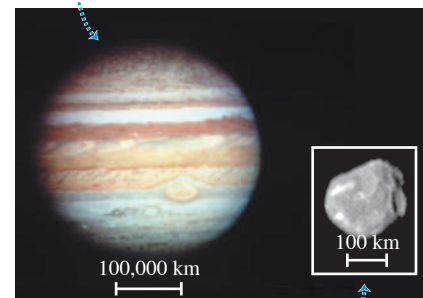
magnitude of force from the other sphere. We use Newton's law of gravitation, Eq. (13.1), to determine F_g :

$$\begin{aligned} F_g &= \frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(0.0100 \text{ kg})(0.500 \text{ kg})}{(0.0500 \text{ m})^2} \\ &= 1.33 \times 10^{-10} \text{ N} \end{aligned}$$

Continued

Figure 13.3 Spherical and nonspherical objects: the planet Jupiter and one of Jupiter's small moons, Amalthea.

Jupiter's mass is very large ($1.90 \times 10^{27} \text{ kg}$), so the mutual gravitational attraction of its parts has pulled it into a nearly spherical shape.



Amalthea, one of Jupiter's small moons, has a relatively tiny mass ($7.17 \times 10^{18} \text{ kg}$, only about 3.8×10^{-9} the mass of Jupiter) and weak mutual gravitation, so it has an irregular shape.

EVALUATE It's remarkable that such a small force could be measured—or even detected—more than 200 years ago. Only a very massive object such as the earth exerts a gravitational force we can feel.

CAUTION **Newton's third law applies to gravitational forces, too** Even though the large sphere in this example has 50 times the mass of the small sphere, each sphere feels the *same* magnitude of force from the other. But because their masses are different, the accelerations

of the two objects in response to that magnitude of force are different (see the following example). **I**

KEYCONCEPT Any two objects exert attractive gravitational forces on each other that are proportional to the product of the masses of the two objects and inversely proportional to the square of the distance between their centers (Newton's law of gravitation).

EXAMPLE 13.2 Acceleration due to gravitational attraction

Suppose the two spheres in Example 13.1 are placed with their centers 0.0500 m apart at a point in space far removed from all other objects. What is the magnitude of the acceleration of each, relative to an inertial system?

IDENTIFY, SET UP, and EXECUTE Each sphere exerts on the other a gravitational force of the same magnitude F_g , which we found in Example 13.1. We can ignore any other forces. The *acceleration* magnitudes a_1 and a_2 are different because the masses are different. To determine these we'll use Newton's second law:

$$a_1 = \frac{F_g}{m_1} = \frac{1.33 \times 10^{-10} \text{ N}}{0.0100 \text{ kg}} = 1.33 \times 10^{-8} \text{ m/s}^2$$

$$a_2 = \frac{F_g}{m_2} = \frac{1.33 \times 10^{-10} \text{ N}}{0.500 \text{ kg}} = 2.66 \times 10^{-10} \text{ m/s}^2$$

EVALUATE The larger sphere has 50 times the mass of the smaller one and hence has $\frac{1}{50}$ the acceleration. These accelerations are *not* constant; the gravitational forces increase as the spheres move toward each other.

KEYCONCEPT The gravitational forces that two objects exert on each other are always equal in magnitude. The accelerations caused by these forces, however, will be different in magnitude for the two objects if they have different masses.

EXAMPLE 13.3 Superposition of gravitational forces

Many stars belong to *systems* of two or more stars held together by their mutual gravitational attraction. **Figure 13.5** shows a three-star system at an instant when the stars are at the vertices of a 45° right triangle. Find the total gravitational force exerted on the small star by the two large ones.

IDENTIFY, SET UP, and EXECUTE We use the principle of superposition: The total force $\vec{F}$ on the small star is the vector sum of the forces $\vec{F}_1$ and $\vec{F}_2$ due to each large star, as Fig. 13.5 shows. We assume that the stars are spheres as in Fig. 13.2. We first calculate the magnitudes F_1 and F_2 from Eq. (13.1) and then compute the vector sum by using components:

$$F_1 = \frac{\left[\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)}{\times (8.00 \times 10^{30} \text{ kg})(1.00 \times 10^{30} \text{ kg})} \right]}{(2.00 \times 10^{12} \text{ m})^2 + (2.00 \times 10^{12} \text{ m})^2} = 6.67 \times 10^{25} \text{ N}$$

$$F_2 = \frac{\left[\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)}{\times (8.00 \times 10^{30} \text{ kg})(1.00 \times 10^{30} \text{ kg})} \right]}{(2.00 \times 10^{12} \text{ m})^2} = 1.33 \times 10^{26} \text{ N}$$

The x - and y -components of these forces are

$$F_{1x} = (6.67 \times 10^{25} \text{ N})(\cos 45^\circ) = 4.72 \times 10^{25} \text{ N}$$

$$F_{1y} = (6.67 \times 10^{25} \text{ N})(\sin 45^\circ) = 4.72 \times 10^{25} \text{ N}$$

$$F_{2x} = 1.33 \times 10^{26} \text{ N}$$

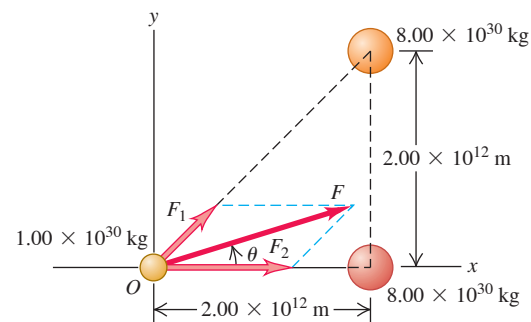
$$F_{2y} = 0$$

The components of the total force $\vec{F}$ on the small star are

$$F_x = F_{1x} + F_{2x} = 1.81 \times 10^{26} \text{ N}$$

$$F_y = F_{1y} + F_{2y} = 4.72 \times 10^{25} \text{ N}$$

Figure 13.5 The total gravitational force on the small star (at O) is the vector sum of the forces exerted on it by the two larger stars. (For comparison, the mass of the sun—a rather ordinary star—is $1.99 \times 10^{30} \text{ kg}$ and the earth–sun distance is $1.50 \times 10^{11} \text{ m}$.)



The magnitude of $\vec{F}$ and its angle θ (see Fig. 13.5) are

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(1.81 \times 10^{26} \text{ N})^2 + (4.72 \times 10^{25} \text{ N})^2} = 1.87 \times 10^{26} \text{ N}$$

$$\theta = \arctan \frac{F_y}{F_x} = \arctan \frac{4.72 \times 10^{25} \text{ N}}{1.81 \times 10^{26} \text{ N}} = 14.6^\circ$$

EVALUATE While the force magnitude F is tremendous, the magnitude of the resulting acceleration is not: $a = F/m = (1.87 \times 10^{26} \text{ N})/(1.00 \times 10^{30} \text{ kg}) = 1.87 \times 10^{-4} \text{ m/s}^2$. Furthermore, the force $\vec{F}$ is *not* directed toward the center of mass of the two large stars.

KEYCONCEPT When there are two objects that exert gravitational forces on a third object, use vector addition to find the magnitude and direction of the net gravitational force on the third object.

Why Gravitational Forces Are Important

Comparing Examples 13.1 and 13.3 shows that gravitational forces are negligible between ordinary household-sized objects but very substantial between objects that are the size of stars. Indeed, gravitation is *the* most important force on the scale of planets, stars, and galaxies (**Fig. 13.6**). It is responsible for holding our earth together and for keeping the planets in orbit about the sun. The mutual gravitational attraction between different parts of the sun compresses material at the sun's core to very high densities and temperatures, making it possible for nuclear reactions to take place there. These reactions generate the sun's energy output, which makes it possible for life to exist on earth and for you to read these words.

The gravitational force is so important on the cosmic scale because it acts *at a distance*, without any direct contact between objects. Electric and magnetic forces have this same remarkable property, but they are less important on astronomical scales because large accumulations of matter are electrically neutral; that is, they contain equal amounts of positive and negative charge. As a result, the electric and magnetic forces between stars or planets are very small or zero. The strong and weak interactions that we discussed in Section 5.5 also act at a distance, but their influence is negligible at distances much greater than the diameter of an atomic nucleus (about 10^{-14} m).

A useful way to describe forces that act at a distance is in terms of a *field*. One object sets up a disturbance or field at all points in space, and the force that acts on a second object at a particular point is its response to the first object's field at that point. There is a field associated with each force that acts at a distance, and so we refer to gravitational fields, electric fields, magnetic fields, and so on. We won't need the field concept for our study of gravitation in this chapter, so we won't discuss it further here. But in later chapters we'll find that the field concept is an extraordinarily powerful tool for describing electric and magnetic interactions.

TEST YOUR UNDERSTANDING OF SECTION 13.1 The planet Saturn has about 100 times the mass of the earth and is about 10 times farther from the sun than the earth is. Compared to the acceleration of the earth caused by the sun's gravitational pull, how great is the acceleration of Saturn due to the sun's gravitation? (i) 100 times greater; (ii) 10 times greater; (iii) the same; (iv) $\frac{1}{10}$ as great; (v) $\frac{1}{100}$ as great.

ANSWER (v) From Eq. (13.1), the gravitational force of the sun (mass m_1) on a planet (mass m_2) at a distance r away has magnitude $F_g = Gm_1m_2/r^2$. Compared to the earth, Saturn has a value of r that is $10^2 = 100$ times greater and a value of m_2 that is also 100 times greater. Hence the force that the sun exerts on Saturn has the same magnitude as the force that the sun exerts on earth. The acceleration of a planet equals the net force divided by the planet's mass: Since Saturn has 100 times more mass than the earth, its acceleration is $\frac{1}{100}$ as great as that of the earth.

13.2 WEIGHT

We defined the *weight* of an object in Section 4.4 as the attractive gravitational force exerted on it by the earth. We can now broaden our definition and say that *the weight of an object is the total gravitational force exerted on the object by all other objects in the universe*. When the object is near the surface of the earth, we can ignore all other gravitational forces and consider the weight as just the earth's gravitational attraction. At the surface of the *moon* we consider an object's weight to be the gravitational attraction of the moon, and so on.

If we again model the earth as a spherically symmetric object with radius R_E , the weight of a small object at the earth's surface (a distance R_E from its center) is

$$\text{Weight of an object at the earth's surface ...} \quad w = F_g = \frac{Gm_E m}{R_E^2} \quad (13.3)$$

... equals gravitational force the earth exerts on object.

Gravitational constant G Mass of the earth m_E Mass of object m Radius of the earth R_E

Figure 13.6 Our solar system is part of a spiral galaxy like this one, which contains roughly 10^{11} stars as well as gas, dust, and other matter. The entire assemblage is held together by the mutual gravitational attraction of all the matter in the galaxy.



APPLICATION Walking and Running on the Moon

Running on the Moon You automatically transition from a walk to a run when the vertical force you exert on the ground—which, by Newton’s third law, equals the vertical force the ground exerts on you—exceeds your weight. This transition from walking to running happens at much lower speeds on the moon, where objects weigh only 17% as much as on earth. Hence, the Apollo astronauts found themselves running even when moving relatively slowly during their moon “walks.”



Figure 13.7 In an airliner at high altitude, you are farther from the center of the earth than when on the ground and hence weigh slightly less. Can you show that at an altitude of 10 km above the surface, you weigh 0.3% less than you do on the ground?



But we also know from Section 4.4 that the weight w of an object is the force that causes the acceleration g of free fall, so by Newton's second law, $w = mg$. Equating this with Eq. (13.3) and dividing by m , we find

$$\text{Acceleration due to gravity at the earth's surface} \quad g = \frac{G m_E}{R_E^2} \quad (13.4)$$

The acceleration due to gravity g is independent of the mass m of the object because m doesn't appear in this equation. We already knew that, but we can now see how it follows from the law of gravitation.

We can *measure* all the quantities in Eq. (13.4) except for m_E , so this relationship allows us to compute the mass of the earth. Solving Eq. (13.4) for m_E and using $R_E = 6370 \text{ km} = 6.37 \times 10^6 \text{ m}$ and $g = 9.80 \text{ m/s}^2$, we find

$$m_E = \frac{gR_E^2}{G} = 5.96 \times 10^{24} \text{ kg}$$

This is very close to the currently accepted value of 5.972×10^{24} kg. Once Cavendish had measured G , he computed the mass of the earth in just this way.

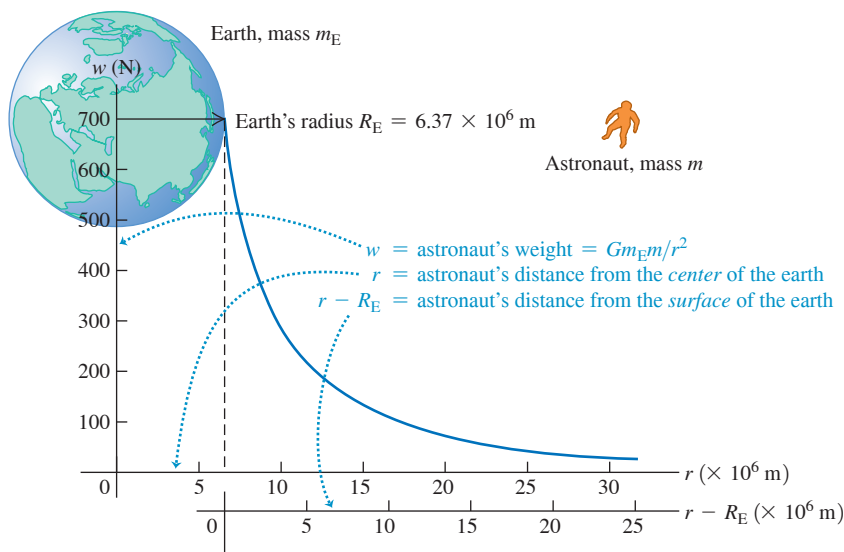
At a point above the earth's surface a distance r from the center of the earth (a distance $r - R_E$ above the surface), the weight of an object is given by Eq. (13.3) with R_E replaced by r :

$$w = F_g = \frac{Gm_E m}{r^2} \quad (13.5)$$

The weight of an object decreases inversely with the square of its distance from the earth's center (**Fig. 13.7**). **Figure 13.8** shows how the weight varies with height above the earth for an astronaut who weighs 700 N at the earth's surface.

The *apparent* weight of an object on earth differs slightly from the earth's gravitational force because the earth rotates and is therefore not precisely an inertial frame of reference. We've ignored this relatively small effect in our discussion but will consider it carefully in Section 13.7.

Figure 13.8 An astronaut who weighs 700 N at the earth's surface experiences less gravitational attraction when above the surface. The relevant distance r is from the astronaut to the *center* of the earth (*not* from the astronaut to the earth's surface).



While the earth is an approximately spherically symmetric distribution of mass, it is *not* uniform throughout its volume. To demonstrate this, let's first calculate the average *density*, or mass per unit volume, of the earth. If we assume a spherical earth, the volume is

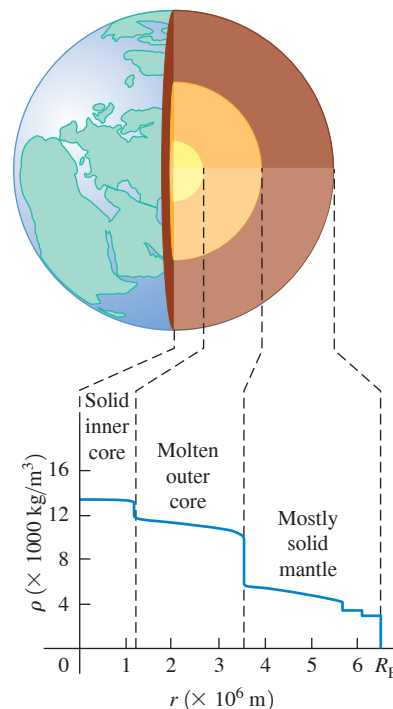
$$\begin{aligned} V_E &= \frac{4}{3}\pi R_E^3 = \frac{4}{3}\pi(6.37 \times 10^6 \text{ m})^3 \\ &= 1.08 \times 10^{21} \text{ m}^3 \end{aligned}$$

The average density ρ (the Greek letter rho) of the earth is the total mass divided by the total volume:

$$\begin{aligned} \rho &= \frac{m_E}{V_E} = \frac{5.97 \times 10^{24} \text{ kg}}{1.08 \times 10^{21} \text{ m}^3} \\ &= 5500 \text{ kg/m}^3 = 5.5 \text{ g/cm}^3 \end{aligned}$$

(Compare to the density of water, $1000 \text{ kg/m}^3 = 1.00 \text{ g/cm}^3$.) If the earth were uniform, rocks near the earth's surface would have this same density. In fact, the density of surface rocks is substantially lower, ranging from about 2000 kg/m^3 for sedimentary rocks to about 3300 kg/m^3 for basalt. So the earth *cannot* be uniform, and its interior must be much more dense than its surface in order that the *average* density be 5500 kg/m^3 . According to geophysical models of the earth's interior, the maximum density at the center is about $13,000 \text{ kg/m}^3$. **Figure 13.9** is a graph of density as a function of distance from the center.

Figure 13.9 The density ρ of the earth decreases with increasing distance r from its center.



EXAMPLE 13.4 Gravity on Mars

A robotic lander with an earth weight of 3430 N is sent to Mars, which has radius $R_M = 3.39 \times 10^6 \text{ m}$ and mass $m_M = 6.42 \times 10^{23} \text{ kg}$ (see Appendix F). Find the weight F_g of the lander on the Martian surface and the acceleration there due to gravity, g_M .

IDENTIFY and SET UP To find F_g we use Eq. (13.3), replacing m_E and R_E with m_M and R_M . We determine the lander mass m from the lander's earth weight w and then find g_M from $F_g = mg_M$.

EXECUTE The lander's earth weight is $w = mg$, so

$$m = \frac{w}{g} = \frac{3430 \text{ N}}{9.80 \text{ m/s}^2} = 350 \text{ kg}$$

The mass is the same no matter where the lander is. From Eq. (13.3), the lander's weight on Mars is

$$\begin{aligned} F_g &= \frac{Gm_M m}{R_M^2} \\ &= \frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(6.42 \times 10^{23} \text{ kg})(350 \text{ kg})}{(3.39 \times 10^6 \text{ m})^2} \\ &= 1.30 \times 10^3 \text{ N} \end{aligned}$$

The acceleration due to gravity on Mars is

$$g_M = \frac{F_g}{m} = \frac{1.30 \times 10^3 \text{ N}}{350 \text{ kg}} = 3.7 \text{ m/s}^2$$

EVALUATE Even though Mars has just 11% of the earth's mass ($6.42 \times 10^{23} \text{ kg}$ versus $5.97 \times 10^{24} \text{ kg}$), the acceleration due to gravity g_M (and hence an object's weight F_g) is roughly 40% as large as on earth. That's because g_M is also inversely proportional to the square of the planet's radius, and Mars has only 53% the radius of earth ($3.39 \times 10^6 \text{ m}$ versus $6.37 \times 10^6 \text{ m}$).

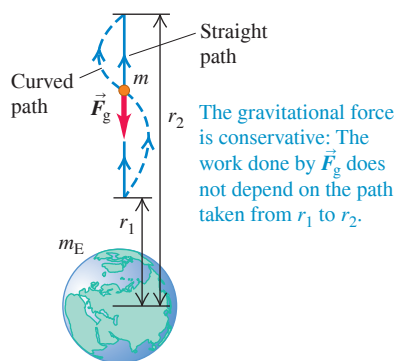
You can check our result for g_M by using Eq. (13.4), with appropriate replacements. Do you get the same answer?

KEYCONCEPT An object's weight at the surface of a planet equals the gravitational force exerted on the object by that planet. The distance between the planet and the object equals the planet's radius. The acceleration due to gravity at a planet's surface is proportional to the planet's mass and inversely proportional to the square of its radius.

TEST YOUR UNDERSTANDING OF SECTION 13.2 Rank the following hypothetical planets in order from highest to lowest value of g at the surface. (Here m_P and R_P are the mass and radius of the planet, and m_E and R_E are the mass and radius of the earth.)

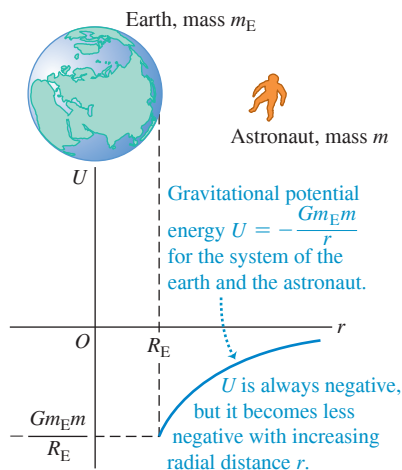
- $m_P = 2m_E$, $R_P = 2R_E$;
- $m_P = 4m_E$, $R_P = 4R_E$;
- $m_P = 4m_E$, $R_P = 2R_E$;
- $m_P = 2m_E$, $R_P = 4R_E$.

Figure 13.10 Calculating the work done on an object by the gravitational force as the object moves from radial coordinate r_1 to r_2 .



CAUTION **Gravitational force vs. gravitational potential energy** Don't confuse the expressions for gravitational force, Eq. (13.7), and gravitational potential energy, Eq. (13.9). The force F_r is proportional to $1/r^2$, while potential energy U is proportional to $1/r$.

Figure 13.11 A graph of the gravitational potential energy U for the system of the earth (mass m_E) and an astronaut (mass m) versus the astronaut's distance r from the center of the earth.



ANSWER (iii) $4/2^2 = 1$ time as great—that is, the same as on earth; and (iv) $2/2^2 = 1/4$ as great. (iii) $4/2^2 = 1$ time as great; (ii) $2/2^2 = 1/4$ as great; (i) $2/2^2 = 1/4$ as great. It follows that compared to the value of g at the earth's surface, the value of g_p on each planet is (i) $2/2^2 = 1/4$ as great; (ii) $4/4^2 = 1/4$ as great; (iii) $4/2^2 = 1$ time as great; and (iv) $2/2^2 = 1/4$ as great. That is, g_p is directly proportional to the planet's mass and inversely proportional to the square of its radius. It follows that compared to the value of g at the earth's surface, the value of g_p on each planet is (i) $2/2^2 = 1/4$ as great; (ii) $4/4^2 = 1/4$ as great; (iii) $4/2^2 = 1$ time as great; and (iv) $2/2^2 = 1/4$ as great. From Eq. (13.4), the acceleration due to gravity at the surface of a planet of mass m_p and radius R_p is $g_p = Gm_p/R_p^2$. That is, g_p is directly proportional to the planet's mass and inversely proportional to the square of its radius. It follows that compared to the value of g at the earth's surface, the value of g_p on each planet is (i) $2/2^2 = 1/4$ as great; (ii) $4/4^2 = 1/4$ as great; (iii) $4/2^2 = 1$ time as great; and (iv) $2/2^2 = 1/4$ as great.

13.3 GRAVITATIONAL POTENTIAL ENERGY

When we first introduced gravitational potential energy in Section 7.1, we assumed that the earth's gravitational force on an object of mass m doesn't depend on the object's height. This led to the expression $U = mgy$. But Eq. (13.2), $F_g = Gm_Em/r^2$, shows that the gravitational force exerted by the earth (mass m_E) *does* in general depend on the distance r from the object to the earth's center. For problems in which an object can be far from the earth's surface, we need a more general expression for gravitational potential energy.

To find this expression, we follow the same steps as in Section 7.1. We consider an object of mass m outside the earth, and first compute the work W_{grav} done by the gravitational force when the object moves directly away from or toward the center of the earth from $r = r_1$ to $r = r_2$, as in **Fig. 13.10**. This work is given by

$$W_{\text{grav}} = \int_{r_1}^{r_2} F_r dr \quad (13.6)$$

where F_r is the radial component of the gravitational force $\vec{F}$ —that is, the component in the direction *outward* from the center of the earth. Because $\vec{F}$ points directly *inward* toward the center of the earth, F_r is negative. It differs from Eq. (13.2), the magnitude of the gravitational force, by a minus sign:

$$F_r = -\frac{Gm_Em}{r^2} \quad (13.7)$$

Substituting Eq. (13.7) into Eq. (13.6), we see that W_{grav} is given by

$$W_{\text{grav}} = -Gm_Em \int_{r_1}^{r_2} \frac{dr}{r^2} = \frac{Gm_Em}{r_2} - \frac{Gm_Em}{r_1} \quad (13.8)$$

The path doesn't have to be a straight line; it could also be a curve like the one in Fig. 13.10. By an argument similar to that in Section 7.1, this work depends on only the initial and final values of r , not on the path taken. This also proves that the gravitational force is always *conservative*.

We now define the corresponding potential energy U so that $W_{\text{grav}} = U_1 - U_2$, as in Eq. (7.3). Comparing this with Eq. (13.8), we see that the appropriate definition for **gravitational potential energy** is

$$U = -\frac{Gm_Em}{r} \quad (13.9)$$

Gravitational constant G Mass of the earth m_E Mass of object m Distance of object from the earth's center r

Gravitational potential energy (general expression)

Figure 13.11 shows how the gravitational potential energy depends on the distance r between the object of mass m and the center of the earth. When the object moves away from the earth, r increases, the gravitational force does negative work, and U increases (i.e., becomes less negative). When the object “falls” toward earth, r decreases, the gravitational work is positive, and the potential energy decreases (i.e., becomes more negative).

CAUTION **Don't worry about gravitational potential energy being negative** You may be troubled by Eq. (13.9) because it states that gravitational potential energy is always negative. But in fact you've seen negative values of U before. In using the formula $U = mgy$ in Section 7.1, we found that U was negative whenever the object of mass m was at a value of y below the arbitrary height we chose to be $y = 0$ —that is, whenever the object and the earth were closer together than some

arbitrary distance. (See, for instance, Example 7.2 in Section 7.1.) In defining U by Eq. (13.9), we have chosen U to be zero when the object of mass m is infinitely far from the earth ($r = \infty$). As the object moves toward the earth, gravitational potential energy decreases and so becomes negative. ■

If we wanted, we could make $U = 0$ at the earth's surface, where $r = R_E$, by adding the quantity Gm_Em/R_E to Eq. (13.9). This would make U positive when $r > R_E$. We won't do this for two reasons: One, it would complicate the expression for U ; two, the added term would not affect the *difference* in potential energy between any two points, which is the only physically significant quantity.

If the earth's gravitational force on an object is the only force that does work, then the total mechanical energy of the system of the earth and object is constant, or *conserved*. In the following example we'll use this principle to calculate **escape speed**, the speed required for an object to escape completely from a planet.

EXAMPLE 13.5 “From the earth to the moon”

WITH  **ARIATION PROBLEMS**

In Jules Verne's 1865 story with this title, three men went to the moon in a shell fired from a giant cannon sunk in the earth in Florida. (a) Find the minimum muzzle speed needed to shoot a shell straight up to a height above the earth equal to the earth's radius R_E . (b) Find the minimum muzzle speed that would allow a shell to escape from the earth completely (the *escape speed*). Neglect air resistance, the earth's rotation, and the gravitational pull of the moon. The earth's radius and mass are $R_E = 6.37 \times 10^6$ m and $m_E = 5.97 \times 10^{24}$ kg.

IDENTIFY and SET UP Once the shell leaves the cannon muzzle, only the (conservative) gravitational force does work. Hence we can use conservation of total mechanical energy to find the speed at which the shell must leave the muzzle so as to come to a halt (a) at two earth radii from the earth's center and (b) at an infinite distance from earth. The energy-conservation equation is $K_1 + U_1 = K_2 + U_2$, with U given by Eq. (13.9).

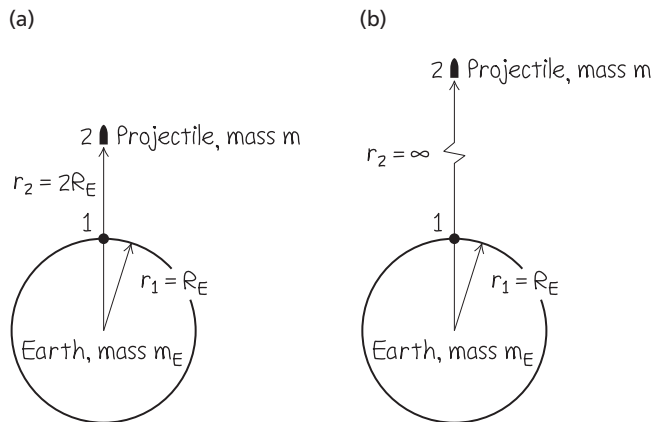
In **Fig. 13.12** point 1 is at $r_1 = R_E$, where the shell leaves the cannon with speed v_1 (the target variable). Point 2 is where the shell reaches its maximum height; in part (a) $r_2 = 2R_E$ (Fig. 13.12a), and in part (b) $r_2 = \infty$ (Fig. 13.12b). In both cases $v_2 = 0$ and $K_2 = 0$. Let m be the mass of the shell (with passengers).

EXECUTE (a) We solve the energy-conservation equation for v_1 :

$$K_1 + U_1 = K_2 + U_2$$

$$\frac{1}{2}mv_1^2 + \left(-\frac{Gm_Em}{R_E}\right) = 0 + \left(-\frac{Gm_Em}{2R_E}\right)$$

Figure 13.12 Our sketches for this problem.



$$v_1 = \sqrt{\frac{Gm_E}{R_E}} = \sqrt{\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}{6.37 \times 10^6 \text{ m}}}$$

$$= 7910 \text{ m/s} (= 28,500 \text{ km/h} = 17,700 \text{ mi/h})$$

(b) Now $r_2 = \infty$ so $U_2 = 0$ (see Fig. 13.11). Since $K_2 = 0$, the total mechanical energy $K_2 + U_2$ is zero in this case. Again we solve the energy-conservation equation for v_1 :

$$\frac{1}{2}mv_1^2 + \left(-\frac{Gm_Em}{R_E}\right) = 0 + 0$$

$$v_1 = \sqrt{\frac{2Gm_E}{R_E}}$$

$$= \sqrt{\frac{2(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}{6.37 \times 10^6 \text{ m}}}$$

$$= 1.12 \times 10^4 \text{ m/s} (= 40,200 \text{ km/h} = 25,000 \text{ mi/h})$$

EVALUATE Our results don't depend on the mass of the shell or the direction of launch. A modern spacecraft launched from Florida must attain essentially the speed found in part (b) to escape the earth; however, before launch it's already moving at 410 m/s to the east because of the earth's rotation. Launching to the east takes advantage of this “free” contribution toward escape speed.

To generalize, the initial speed v_1 needed for an object to escape from the surface of a spherical object of mass M and radius R (ignoring air resistance) is $v_1 = \sqrt{2GM/R}$ (escape speed). This equation yields 5.03×10^3 m/s for Mars, 6.02×10^4 m/s for Jupiter, and 6.18×10^5 m/s for the sun.

If the initial speed is greater than the speed $v_1 = \sqrt{GM_E/R_E}$ that we found in part (a), the shell will still be moving when it gets to $r_2 = 2R_E$. In the same way, if the initial speed is greater than the escape speed $v_1 = \sqrt{2GM_E/R_E}$ that we found in part (b), the shell will still be moving even when it is very far from the earth (at $r_2 = \infty$).

KEYCONCEPT When an object moves far away from a planet's surface, use Eq. (13.11) for the gravitational potential energy. The potential energy for the system of planet and object increases as the object moves farther away; it reaches its maximum value of zero when the object is infinitely far away.

More on Gravitational Potential Energy

As a final note, let's show that when we are close to the earth's surface, Eq. (13.9) reduces to the familiar $U = mgy$ from Chapter 7. We first rewrite Eq. (13.8) as

$$W_{\text{grav}} = Gm_E m \frac{r_1 - r_2}{r_1 r_2}$$

If the object stays close to the earth, then in the denominator we may replace r_1 and r_2 by R_E , the earth's radius, so

$$W_{\text{grav}} = Gm_E m \frac{r_1 - r_2}{R_E^2}$$

According to Eq. (13.4), $g = Gm_E/R_E^2$, so

$$W_{\text{grav}} = mg(r_1 - r_2)$$

If we replace the r 's by y 's, this is just Eq. (7.1) for the work done by a constant gravitational force. In Section 7.1 we used this equation to derive Eq. (7.2), $U = mgy$, so we may consider Eq. (7.2) for gravitational potential energy to be a special case of the more general Eq. (13.9).

TEST YOUR UNDERSTANDING OF SECTION 13.3 If a planet has the same surface gravity as the earth (that is, the same value of g at the surface), what is its escape speed? (i) The same as the earth's; (ii) less than the earth's; (iii) greater than the earth's; (iv) any of these are possible.

ANSWER (iv) For a planet of mass m_p and radius R_p , the surface gravity is $g = Gm_p/R_p^2$ while the escape speed is $v_{\text{esc}} = \sqrt{2Gm_p/R_p}$. Comparing these two expressions, you get $v_{\text{esc}} = \sqrt{2gR_p}$. So even if a planet has the same value of g as the earth, its escape speed can be different, depending on how its radius R_p compares with the earth's radius. For the planet Saturn, for example, m_p is about 100 times the earth's mass and R_p is about 10 times the earth's radius. The value of g is different than on earth by a factor of $(100)/(10)^2 = 1$ (i.e., it is the same as on earth), while the escape speed is greater by a factor of $\sqrt{100/10} = 3.2$.

Figure 13.13 With a mass of approximately 4.5×10^5 kg and a width of over 108 m, the International Space Station is the largest satellite ever placed in orbit.



13.4 THE MOTION OF SATELLITES

Artificial satellites orbiting the earth are a familiar part of technology (**Fig. 13.13**). But how do they stay in orbit, and what determines the properties of their orbits? We can use Newton's laws and the law of gravitation to provide the answers. In the next section we'll analyze the motion of planets in the same way.

To begin, think back to the discussion of projectile motion in Section 3.3. In Example 3.6 a motorcycle rider rides horizontally off the edge of a cliff, launching himself into a parabolic path that ends on the flat ground at the base of the cliff. If he survives and repeats the experiment with increased launch speed, he will land farther from the starting point. We can imagine him launching himself with great enough speed that the earth's curvature becomes significant. As he falls, the earth curves away beneath him. If he is going fast enough, and if his launch point is high enough that he clears the mountaintops, he may be able to go right on around the earth without ever landing.

Figure 13.14 shows a variation on this theme. We launch a projectile from point A in the direction AB , tangent to the earth's surface. Trajectories 1 through 7 show the effect of increasing the initial speed. In trajectories 3 through 5 the projectile misses the earth and becomes a satellite. If there is no retarding force such as air resistance, the projectile's speed when it returns to point A is the same as its initial speed and it repeats its motion indefinitely.

Trajectories 1 through 5 close on themselves and are called **closed orbits**. All closed orbits are ellipses or segments of ellipses; trajectory 4 is a circle, a special case of an ellipse. (We'll discuss the properties of an ellipse in Section 13.5.) Trajectories 6 and 7 are **open orbits**. For these paths the projectile never returns to its starting point but travels ever farther away from the earth.

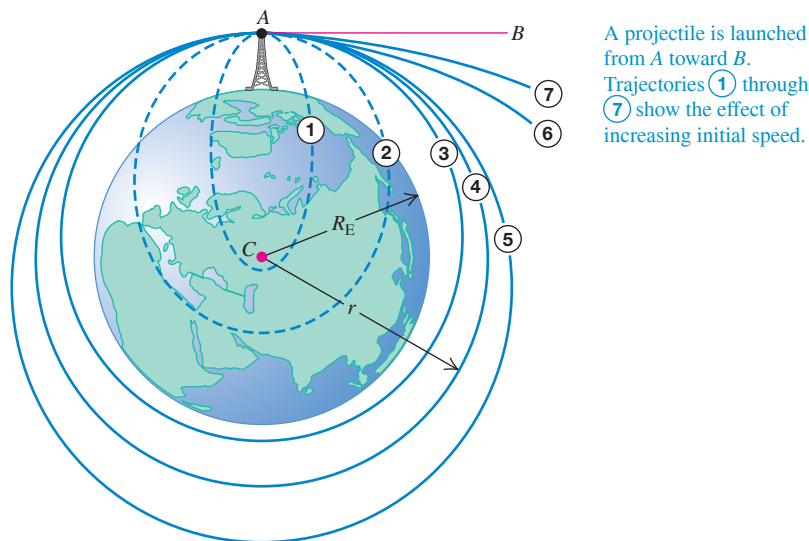


Figure 13.14 Trajectories of a projectile launched from a great height (ignoring air resistance). Orbits 1 and 2 would be completed as shown if the earth were a point mass at C. (This illustration is based on one in Isaac Newton's *Principia*.)

Satellites: Circular Orbits

A *circular orbit*, like trajectory 4 in Fig. 13.14, is the simplest case. It is also an important case, since many artificial satellites have nearly circular orbits and the orbits of the planets around the sun are also fairly circular. The only force acting on a satellite in circular orbit around the earth is the earth's gravitational attraction, which is directed toward the center of the earth and hence toward the center of the orbit (Fig. 13.15). As we discussed in Section 5.4, this means that the satellite is in *uniform circular motion* and its speed is constant. The satellite isn't falling *toward* the earth; rather, it's constantly falling *around* the earth. In a circular orbit the speed is just right to keep the distance from the satellite to the center of the earth constant.

Let's see how to find the constant speed v of a satellite in a circular orbit. The radius of the orbit is r , measured from the *center* of the earth; the acceleration of the satellite has magnitude $a_{\text{rad}} = v^2/r$ and is always directed toward the center of the circle. By the law of gravitation, the net force (gravitational force) on the satellite of mass m has magnitude $F_g = Gm_E m/r^2$ and is in the same direction as the acceleration. Newton's second law ($\Sigma \vec{F} = m\vec{a}$) then tells us that

$$\frac{Gm_E m}{r^2} = \frac{mv^2}{r}$$

Solving this for v , we find

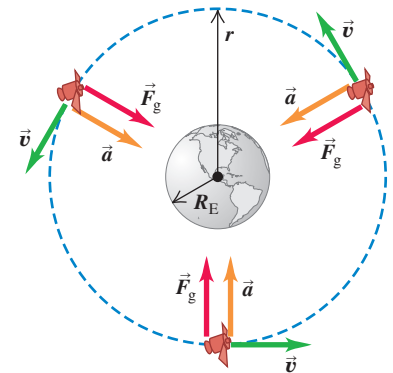
$$v = \sqrt{\frac{Gm_E}{r}} \quad (13.10)$$

Speed of satellite in a circular orbit around the earth Gravitational constant Mass of the earth Radius of orbit

This relationship shows that we can't choose the orbit radius r and the speed v independently; for a given radius r , the speed v for a circular orbit is determined.

The satellite's mass m doesn't appear in Eq. (13.10), which shows that the motion of a satellite does not depend on its mass. An astronaut on board an orbiting space station is herself a satellite of the earth, held by the earth's gravity in the same orbit as the station. The astronaut has the same velocity and acceleration as the station, so nothing is pushing her against the station's floor or walls. She is in a state of *apparent weightlessness*, as in a freely falling elevator; see the discussion following Example 5.9 in Section 5.2. (*True* weightlessness would occur only if the astronaut were infinitely far from any other masses, so that the gravitational force on her would be zero.) Indeed, every part of her body is apparently weightless; she feels nothing pushing her stomach against her intestines or her head against her shoulders (Fig. 13.16).

Figure 13.15 The force $\vec{F}_g$ due to the earth's gravitational attraction provides the centripetal acceleration that keeps a satellite in orbit. Compare to Fig. 5.28.



The satellite is in a circular orbit: Its acceleration $\vec{a}$ is always perpendicular to its velocity $\vec{v}$, so its speed v is constant.

Figure 13.16 These astronauts are in a state of apparent weightlessness. Who is right side up?



Apparent weightlessness is not just a feature of circular orbits; it occurs whenever gravity is the only force acting on a spacecraft. Hence it occurs for orbits of any shape, including open orbits such as trajectories 6 and 7 in Fig. 13.14.

We can derive a relationship between the radius r of a circular orbit and the period T , the time for one revolution. The speed v is the distance $2\pi r$ traveled in one revolution, divided by the period:

$$v = \frac{2\pi r}{T} \quad (13.11)$$

We solve Eq. (13.11) for T and substitute v from Eq. (13.10):

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{Gm_E}} = \frac{2\pi r^{3/2}}{\sqrt{Gm_E}} \quad (13.12)$$

Period of a circular orbit around the earth $\rightarrow$ T $\leftarrow$ Orbital speed v $\leftarrow$ Radius of orbit r $\leftarrow$ Gravitational constant G $\leftarrow$ Mass of the earth m_E

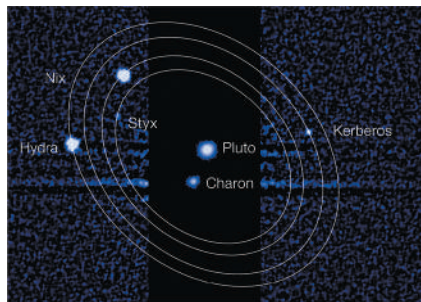
Equations (13.10) and (13.12) show that larger orbits correspond to slower speeds and longer periods. As an example, the International Space Station (Fig. 13.13) orbits 6800 km from the center of the earth (400 km above the earth's surface) with an orbital speed of 7.7 km/s and an orbital period of 93 min. The moon orbits the earth in a much larger orbit of radius 384,000 km, and so has a much slower orbital speed (1.0 km/s) and a much longer orbital period (27.3 days).

It's interesting to compare Eq. (13.10) to the calculation of escape speed in Example 13.5. We see that the escape speed from a spherical object with radius R is $\sqrt{2}$ times greater than the speed of a satellite in a circular orbit at that radius. If our spacecraft is in circular orbit around *any* planet, we have to multiply our speed by a factor of $\sqrt{2}$ to escape to infinity, regardless of the planet's mass.

Since the speed v in a circular orbit is determined by Eq. (13.10) for a given orbit radius r , the total mechanical energy $E = K + U$ is determined as well. Using Eqs. (13.9) and (13.10), we have

$$\begin{aligned} E &= K + U = \frac{1}{2}mv^2 + \left(-\frac{Gm_Em}{r}\right) \\ &= \frac{1}{2}m\left(\frac{Gm_E}{r}\right) - \frac{Gm_Em}{r} \\ &= -\frac{Gm_Em}{2r} \quad (\text{circular orbit}) \end{aligned} \quad (13.13)$$

Figure 13.17 The dwarf planet Pluto is known to have at least five satellites. In accordance with Eqs. (13.10) and (13.12), the larger the satellite's orbit around Pluto, the slower the satellite's orbital speed and the longer its orbital period.



The total mechanical energy in a circular orbit is negative and equal to one-half the potential energy. Increasing the orbit radius r means increasing the total mechanical energy (that is, making E less negative). If the satellite is in a relatively low orbit that encounters the outer fringes of earth's atmosphere, the total mechanical energy decreases due to negative work done by the force of air resistance; as a result, the orbit radius decreases until the satellite hits the ground or burns up in the atmosphere.

We have talked mostly about earth satellites, but we can apply the same analysis to the circular motion of *any* object under its gravitational attraction to a stationary object. **Figure 13.17** shows an example.

EXAMPLE 13.6 A satellite orbit

WITH VARIATION PROBLEMS

You wish to put a 1000 kg satellite into a circular orbit 340 km above the earth's surface. (a) What speed, period, and radial acceleration will it have? (b) How much work must be done to the satellite to put it in orbit? (c) How much additional work would have to be done to make the satellite escape the earth? The earth's radius and mass are given in Example 13.5 (Section 13.3).

IDENTIFY and SET UP The satellite is in a circular orbit, so we can use the equations derived in this section. In part (a), we first find the radius r of the satellite's orbit from its altitude. We then calculate the speed v and period T from Eqs. (13.10) and (13.12) and the acceleration from $a_{\text{rad}} = v^2/r$. In parts (b) and (c), the work required is the difference between the initial and final total mechanical energies, which for a circular orbit is given by Eq. (13.13).

EXECUTE (a) The radius of the satellite's orbit is $r = 6370 \text{ km} + 340 \text{ km} = 6710 \text{ km} = 6.71 \times 10^6 \text{ m}$. From Eq. (13.10), the orbital speed is

$$v = \sqrt{\frac{Gm_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}{6.71 \times 10^6 \text{ m}}} = 7700 \text{ m/s}$$

We find the orbital period from Eq. (13.12):

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.71 \times 10^6 \text{ m})}{7700 \text{ m/s}} = 5470 \text{ s} = 91.2 \text{ min}$$

Finally, the radial acceleration is

$$a_{\text{rad}} = \frac{v^2}{r} = \frac{(7700 \text{ m/s})^2}{6.71 \times 10^6 \text{ m}} = 8.84 \text{ m/s}^2$$

This is the value of g at a height of 340 km above the earth's surface; it is about 10% less than the value of g at the surface.

(b) The work required is the difference between E_2 , the total mechanical energy when the satellite is in orbit, and E_1 , the total mechanical energy when the satellite was at rest on the launch pad. From Eq. (13.13), the energy in orbit is

$$\begin{aligned} E_2 &= -\frac{Gm_E m}{2r} \\ &= -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})(1000 \text{ kg})}{2(6.71 \times 10^6 \text{ m})} \\ &= -2.97 \times 10^{10} \text{ J} \end{aligned}$$

The satellite's kinetic energy is zero on the launch pad ($r = R_E$), so

$$\begin{aligned} E_1 &= K_1 + U_1 = 0 + \left(-\frac{Gm_E m}{R_E}\right) \\ &= -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})(1000 \text{ kg})}{6.37 \times 10^6 \text{ m}} \\ &= -6.25 \times 10^{10} \text{ J} \end{aligned}$$

Hence the work required is

$$\begin{aligned} W_{\text{required}} &= E_2 - E_1 = (-2.97 \times 10^{10} \text{ J}) - (-6.25 \times 10^{10} \text{ J}) \\ &= 3.28 \times 10^{10} \text{ J} \end{aligned}$$

(c) We saw in part (b) of Example 13.5 that the minimum total mechanical energy for a satellite to escape to infinity is zero. Here, the total mechanical energy in the circular orbit is $E_2 = -2.97 \times 10^{10} \text{ J}$; to increase this to zero, an amount of work equal to $2.97 \times 10^{10} \text{ J}$ would have to be done on the satellite, presumably by rocket engines attached to it.

EVALUATE In part (b) we ignored the satellite's initial kinetic energy (while it was still on the launch pad) due to the rotation of the earth. How much difference does this make? (See Example 13.5 for useful data.)

KEYCONCEPT The larger the radius of a satellite's orbit, the greater (less negative) the total mechanical energy, the greater (less negative) the potential energy, and the smaller the kinetic energy (the satellite moves more slowly in a larger orbit). The radius of a satellite's orbit is always measured from the center of the earth.

TEST YOUR UNDERSTANDING OF SECTION 13.4 Your personal spacecraft is in a low-altitude circular orbit around the earth. Air resistance from the outer regions of the atmosphere does negative work on the spacecraft, causing the radius of the circular orbit to decrease slightly. Does the speed of the spacecraft (i) remain the same, (ii) increase, or (iii) decrease?

ANSWER (i) Equation (13.10) shows that in a smaller-radius orbit, the spacecraft has a faster speed. The negative work done by air resistance decreases the total mechanical energy $E = K + U$; the kinetic energy K increases (becomes more positive), but the gravitational potential energy U decreases (becomes more negative) by a greater amount.

13.5 KEPLER'S LAWS AND THE MOTION OF PLANETS

The name *planet* comes from a Greek word meaning “wanderer,” and indeed the planets continuously change their positions in the sky relative to the background of stars. One of the great intellectual accomplishments of the 16th and 17th centuries was the threefold realization that the earth is also a planet, that all planets orbit the sun, and that the apparent motions of the planets as seen from the earth can be used to determine their orbits precisely.

The first and second of these ideas were published by Nicolaus Copernicus in Poland in 1543. The nature of planetary orbits was deduced between 1601 and 1619 by the German astronomer and mathematician Johannes Kepler, using precise data on apparent planetary motions compiled by his mentor, the Danish astronomer Tycho Brahe. By trial and error, Kepler discovered three empirical laws that accurately described the motions of the planets:

KEPLER'S LAWS

1. Each planet moves in an elliptical orbit, with the sun at one focus of the ellipse.
2. A line from the sun to a given planet sweeps out equal areas in equal times.
3. The periods of the planets are proportional to the $\frac{3}{2}$ powers of the major axis lengths of their orbits.

Figure 13.18 Geometry of an ellipse. The sum of the distances SP and $S'P$ is the same for every point on the curve. The sizes of the sun (S) and planet (P) are exaggerated for clarity.

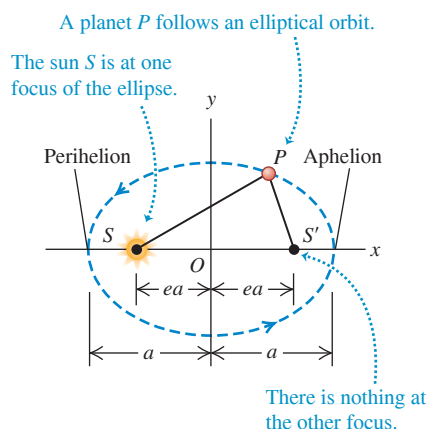
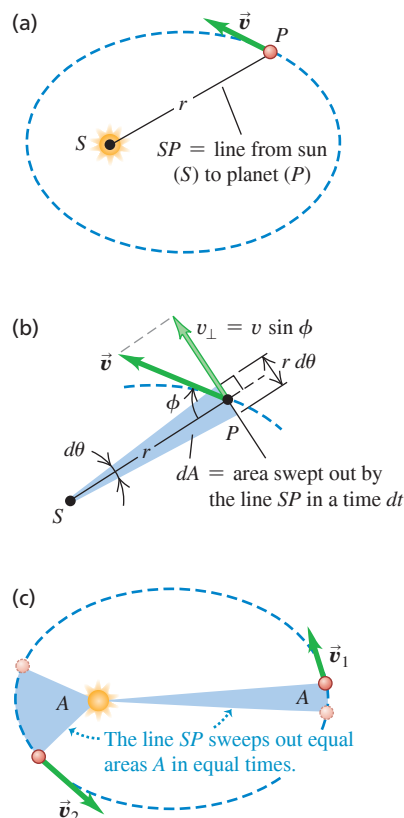


Figure 13.19 (a) The planet (P) moves about the sun (S) in an elliptical orbit. (b) In a time dt the line SP sweeps out an area $dA = \frac{1}{2}(r d\theta)r = \frac{1}{2}r^2 d\theta$. (c) The planet's speed varies so that the line SP sweeps out the same area A in a given time t regardless of the planet's position in its orbit.



Kepler did not know *why* the planets moved in this way. Three generations later, when Newton turned his attention to the motion of the planets, he discovered that each of Kepler's laws can be *derived*; they are consequences of Newton's laws of motion and the law of gravitation. Let's see how each of Kepler's laws arises.

Kepler's First Law

First consider the elliptical orbits described in Kepler's first law. **Figure 13.18** shows the geometry of an ellipse. The longest dimension is the *major axis*, with half-length a ; this half-length is called the **semi-major axis**. The sum of the distances from S to P and from S' to P is the same for all points on the curve. S and S' are the *foci* (plural of *focus*). The sun is at S (not at the center of the ellipse) and the planet is at P ; we think of both as points because the size of each is very small in comparison to the distance between them. There is nothing at the other focus, S' .

The distance of each focus from the center of the ellipse is ea , where e is a dimensionless number between 0 and 1 called the **eccentricity**. If $e = 0$, the two foci coincide and the ellipse is a circle. The actual orbits of the planets are fairly circular; their eccentricities range from 0.007 for Venus to 0.206 for Mercury. (The earth's orbit has $e = 0.017$.) The point in the planet's orbit closest to the sun is the *perihelion*, and the point most distant is the *aphelion*.

Newton showed that for an object acted on by an attractive force proportional to $1/r^2$, the only possible closed orbits are a circle or an ellipse; he also showed that open orbits (trajectories 6 and 7 in Fig. 13.14) must be parabolas or hyperbolas. These results can be derived from Newton's laws and the law of gravitation, together with a lot more differential equations than we're ready for.

Kepler's Second Law

Figure 13.19 shows Kepler's second law. In a small time interval dt , the line from the sun S to the planet P turns through an angle $d\theta$. The area swept out is the colored triangle with height r , base length $r d\theta$, and area $dA = \frac{1}{2}r^2 d\theta$ in Fig. 13.19b. The rate at which area is swept out, dA/dt , is called the *sector velocity*:

$$\frac{dA}{dt} = \frac{1}{2}r^2 \frac{d\theta}{dt} \quad (13.14)$$

The essence of Kepler's second law is that the sector velocity has the same value at all points in the orbit. When the planet is close to the sun, r is small and $d\theta/dt$ is large; when the planet is far from the sun, r is large and $d\theta/dt$ is small.

To see how Kepler's second law follows from Newton's laws, we express dA/dt in terms of the velocity vector $\vec{v}$ of the planet P . The component of $\vec{v}$ perpendicular to the radial line is $v_{\perp} = v \sin \phi$. From Fig. 13.19b the displacement along the direction of $v_{\perp}$ during time dt is $r d\theta$, so we also have $v_{\perp} = r d\theta/dt$. Using this relationship in Eq. (13.14), we find

$$\frac{dA}{dt} = \frac{1}{2}rv \sin \phi \quad (\text{sector velocity}) \quad (13.15)$$

Now $rv \sin \phi$ is the magnitude of the vector product $\vec{r} \times \vec{v}$, which in turn is $1/m$ times the angular momentum $\vec{L} = \vec{r} \times m\vec{v}$ of the planet with respect to the sun. So we have

$$\frac{dA}{dt} = \frac{1}{2m} |\vec{r} \times m\vec{v}| = \frac{L}{2m} \quad (13.16)$$

Thus Kepler's second law—that sector velocity is constant—means that angular momentum is constant!

It is easy to see why the angular momentum of the planet *must* be constant. According to Eq. (10.26), the rate of change of $\vec{L}$ equals the torque of the gravitational force $\vec{F}$ acting on the planet:

$$\frac{d\vec{L}}{dt} = \vec{\tau} = \vec{r} \times \vec{F}$$

In our situation, $\vec{r}$ is the vector from the sun to the planet, and the force $\vec{F}$ is directed from the planet to the sun (**Fig. 13.20**). So these vectors always lie along the same line, and

their vector product $\vec{r} \times \vec{F}$ is zero. Hence $d\vec{L}/dt = \mathbf{0}$. This conclusion does not depend on the $1/r^2$ behavior of the force; angular momentum is conserved for *any* force that acts always along the line joining the particle to a fixed point. Such a force is called a *central force*. (Kepler's first and third laws are valid for a $1/r^2$ force *only*.)

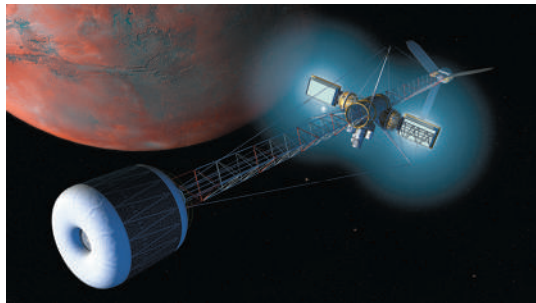
Conservation of angular momentum also explains why the orbit lies in a plane. The vector $\vec{L} = \vec{r} \times m\vec{v}$ is always perpendicular to the plane of the vectors $\vec{r}$ and $\vec{v}$; since $\vec{L}$ is constant in magnitude *and* direction, $\vec{r}$ and $\vec{v}$ always lie in the same plane, which is just the plane of the planet's orbit.

Kepler's Third Law

We have already derived Kepler's third law for the particular case of circular orbits. Equation (13.12) shows that the period of a satellite or planet in a circular orbit is proportional to the $\frac{3}{2}$ power of the orbit radius. Newton was able to show that this same relationship holds for an *elliptical* orbit, with the orbit radius r replaced by the semi-major axis a :

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_S}} \quad (\text{elliptical orbit around the sun}) \quad (13.17)$$

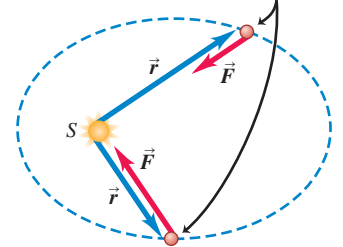
Since the planet orbits the sun, not the earth, we have replaced the earth's mass m_E in Eq. (13.12) with the sun's mass m_S . Note that the period does not depend on the eccentricity e . An asteroid in an elongated elliptical orbit with semi-major axis a will have the same orbital period as a planet in a circular orbit of radius a . The key difference is that the asteroid moves at different speeds at different points in its elliptical orbit (Fig. 13.19c), while the planet's speed is constant around its circular orbit.



BIO APPLICATION Biological Hazards of Interplanetary Travel A spacecraft sent from earth to another planet spends most of its journey coasting along an elliptical orbit with the sun at one focus. Rockets are used at only the start and end of the journey, and even the trip to a nearby planet like Mars takes several months. During its journey, the spacecraft is exposed to cosmic rays—radiation that emanates from elsewhere in our galaxy. (On earth we're shielded from this radiation by our planet's magnetic field, as we'll describe in Chapter 27.) This poses no problem for a robotic spacecraft but would be a severe medical hazard for astronauts undertaking such a voyage.

Figure 13.20 Because the gravitational force that the sun exerts on a planet produces zero torque around the sun, the planet's angular momentum around the sun remains constant.

Same planet at two points in its orbit



- Gravitational force $\vec{F}$ on planet has different magnitudes at different points but is always opposite to vector $\vec{r}$ from sun S to planet.
- Hence $\vec{F}$ produces zero torque around sun.
- Hence angular momentum $\vec{L}$ of planet around sun is constant in both magnitude and direction.

CONCEPTUAL EXAMPLE 13.7 Orbital speeds

WITH VARIATION PROBLEMS

At what point in an elliptical orbit (see Fig. 13.19) does a planet move the fastest? The slowest?

SOLUTION Total mechanical energy is conserved as a planet moves in its orbit. The planet's kinetic energy $K = \frac{1}{2}mv^2$ is maximum when the potential energy $U = -Gm_Sm/r$ is minimum (that is, most negative; see Fig. 13.11), which occurs when the sun–planet distance r is a minimum. Hence the speed v is greatest at perihelion. Similarly, K is minimum when r is maximum, so the speed is slowest at aphelion.

Your intuition about falling objects is helpful here. As the planet falls inward toward the sun, it picks up speed, and its speed is maximum when closest to the sun. The planet slows down as it moves away from the sun, and its speed is minimum at aphelion.

KEYCONCEPT Total mechanical energy remains constant for a planet in an elliptical orbit. The planet's kinetic energy (and speed) therefore increases as it moves closer to the sun and the gravitational potential energy decreases (becomes more negative).

EXAMPLE 13.8 Kepler's third law

WITH VARIATION PROBLEMS

The asteroid Pallas has an orbital period of 4.62 years and an orbital eccentricity of 0.233. Find the semi-major axis of its orbit.

IDENTIFY and SET UP We need Kepler's third law, which relates the period T and the semi-major axis a for an orbiting object (such as an

asteroid). We use Eq. (13.17) to determine a ; from Appendix F we have $m_S = 1.99 \times 10^{30}$ kg, and a conversion factor from Appendix E gives $T = (4.62 \text{ y})(3.156 \times 10^7 \text{ s/y}) = 1.46 \times 10^8 \text{ s}$. Note that we don't need the value of the eccentricity.

Continued

EXECUTE From Eq. (13.17), $a^{3/2} = [(Gm_S)^{1/2}T]/2\pi$. To solve for a , we raise both sides of this expression to the $\frac{2}{3}$ power and then substitute the values of G , m_S , and T :

$$a = \left(\frac{Gm_S T^2}{4\pi^2} \right)^{1/3} = 4.15 \times 10^{11} \text{ m}$$

(Plug in the numbers yourself to check.)

EVALUATE Our result is intermediate between the semi-major axes of Mars and Jupiter (see Appendix F). Most known asteroids orbit in an “asteroid belt” between the orbits of these two planets.

KEYCONCEPT Kepler’s third law allows you to relate the period of an orbit, the semi-major axis of that orbit, and the mass of the object being orbited.

EXAMPLE 13.9 Comet Halley

WITH VARIATION PROBLEMS

Comet Halley (**Fig. 13.21**) moves in an elongated elliptical orbit around the sun. Its distances from the sun at perihelion and aphelion are $8.80 \times 10^7 \text{ km}$ and $5.25 \times 10^9 \text{ km}$, respectively. Find the orbital semi-major axis, eccentricity, and period in years.

IDENTIFY and SET UP We are to find the semi-major axis a , eccentricity e , and orbital period T . We can use Fig. 13.18 to find a and e from the given perihelion and aphelion distances. Knowing a , we can find T from Kepler’s third law, Eq. (13.17).

EXECUTE From Fig. 13.18, the length $2a$ of the major axis equals the sum of the comet–sun distance at perihelion and the comet–sun distance at aphelion. Hence

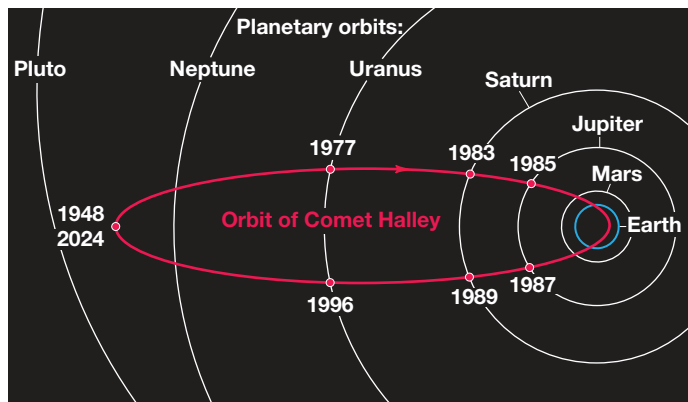
$$a = \frac{(8.80 \times 10^7 \text{ km}) + (5.25 \times 10^9 \text{ km})}{2} = 2.67 \times 10^9 \text{ km}$$

Figure 13.18 also shows that the comet–sun distance at perihelion is $a - ea = a(1 - e)$. This distance is $8.80 \times 10^7 \text{ km}$, so

$$e = 1 - \frac{8.80 \times 10^7 \text{ km}}{a} = 1 - \frac{8.80 \times 10^7 \text{ km}}{2.67 \times 10^9 \text{ km}} = 0.967$$

Figure **13.21** (a) The orbit of Comet Halley. (b) Comet Halley as it appeared in 1986. At the heart of the comet is an icy object, called the nucleus, that is about 10 km across. When the comet’s orbit carries it close to the sun, the heat of sunlight causes the nucleus to partially evaporate. The evaporated material forms the tail, which can be tens of millions of kilometers long.

(a)



(b)



Planetary Motions and the Center of Mass

We have assumed that as a planet or comet orbits the sun, the sun remains absolutely stationary. This can’t be correct; because the sun exerts a gravitational force on the planet, the planet exerts a gravitational force on the sun of the same magnitude but opposite direction. In fact, *both* the sun and the planet orbit around their common

center of mass (**Fig. 13.22**). We've made only a small error by ignoring this effect, however; the sun's mass is about 750 times the total mass of all the planets combined, so the center of mass of the solar system is not far from the center of the sun. Remarkably, astronomers have used this effect to detect the presence of planets orbiting other stars. Sensitive telescopes are able to detect the apparent “wobble” of a star as it orbits the common center of mass of the star and an unseen companion planet. (The planets are too faint to observe directly.) By analyzing these “wobbles,” astronomers have discovered planets in orbit around hundreds of other stars.

The most remarkable result of Newton's analysis of planetary motion is that objects in the heavens obey the *same* laws of motion as do objects on the earth. This *Newtonian synthesis*, as it has come to be called, is one of the great unifying principles of science. It has had profound effects on the way that humanity looks at the universe—not as a realm of impenetrable mystery, but as a direct extension of our everyday world, subject to scientific study and calculation.

TEST YOUR UNDERSTANDING OF SECTION 13.5 The orbit of Comet X has a semi-major axis that is four times longer than the semi-major axis of Comet Y. What is the ratio of the orbital period of X to the orbital period of Y? (i) 2; (ii) 4; (iii) 8; (iv) 16; (v) 32; (vi) 64.

ANSWER

(iii) Equation (13.17) shows that the orbital period T is proportional to the $3/2$ power of the semi-major axis a . Hence the orbital period of Comet X is longer than that of Comet Y by a factor of $4^{3/2} = 8$.

13.6 SPHERICAL MASS DISTRIBUTIONS

We have stated without proof that the gravitational interaction between two spherically symmetric mass distributions is the same as though all the mass of each were concentrated at its center. Now we're ready to prove this statement. Newton searched for a proof for several years, and he delayed publication of the law of gravitation until he found one.

Rather than starting with two spherically symmetric masses, we'll tackle the simpler problem of a point mass m interacting with a thin spherical shell with total mass M . We'll show that when m is outside the sphere, the *potential energy* associated with this gravitational interaction is the same as though M were concentrated in a point at the center of the sphere. We learned in Section 7.4 that the force is the negative derivative of the potential energy, so the *force* on m is also the same as for a point mass M . Our result will also hold for *any* spherically symmetric mass distribution, which we can think of as being made of many concentric spherical shells.

A Point Mass Outside a Spherical Shell

We start by considering a ring on the surface of a shell (**Fig. 13.23a**, next page), centered on the line from the center of the shell to m . We do this because all of the particles that make up the ring are the same distance s from the point mass m . From Eq. (13.9) the potential energy of interaction between the earth (mass m_E) and a point mass m , separated by a distance r , is $U = -Gm_E m/r$. From this expression, we see that the potential energy of interaction between the point mass m and a particle of mass m_i within the ring is

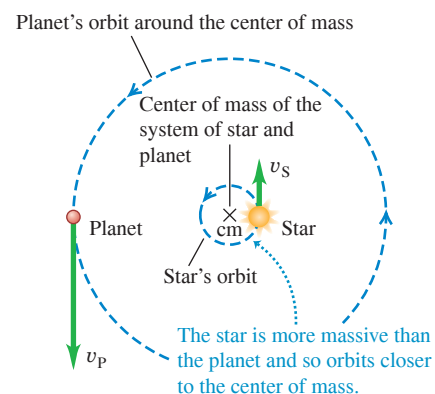
$$U_i = -\frac{Gmm_i}{s}$$

To find the potential energy dU of interaction between m and the entire ring of mass $dM = \sum m_i$, we sum this expression for U_i over all particles in the ring:

$$dU = \sum U_i = \sum \left(-\frac{Gmm_i}{s} \right) = -\frac{Gm}{s} \sum m_i = -\frac{Gm dM}{s} \quad (13.18)$$

To proceed, we need to know the mass dM of the ring. We can find this with the aid of a little geometry. The radius of the shell is R , so in terms of the angle ϕ shown in the figure,

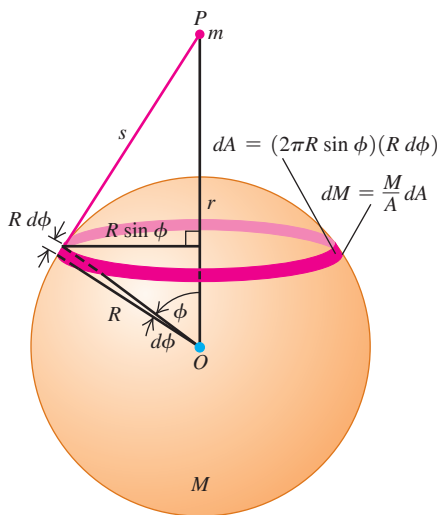
Figure 13.22 Both a star and its planet orbit about their common center of mass.



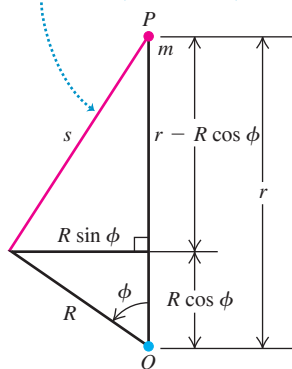
The planet and star are always on opposite sides of the center of mass.

Figure 13.23 Calculating the gravitational potential energy of interaction between a point mass m outside a spherical shell and a ring on the surface of the shell of mass M .

(a) Geometry of the situation



(b) The distance s is the hypotenuse of a right triangle with sides $(r - R \cos \phi)$ and $R \sin \phi$.



the radius of the ring is $R \sin \phi$, and its circumference is $2\pi R \sin \phi$. The width of the ring is $R d\phi$, and its area dA is approximately equal to its width times its circumference:

$$dA = 2\pi R^2 \sin \phi d\phi$$

The ratio of the ring mass dM to the total mass M of the shell is equal to the ratio of the area dA of the ring to the total area $A = 4\pi R^2$ of the shell:

$$\frac{dM}{M} = \frac{2\pi R^2 \sin \phi d\phi}{4\pi R^2} = \frac{1}{2} \sin \phi d\phi \quad (13.19)$$

Now we solve Eq. (13.19) for dM and substitute the result into Eq. (13.18) to find the potential energy of interaction between point mass m and the ring:

$$dU = -\frac{GMm \sin \phi d\phi}{2s} \quad (13.20)$$

The total potential energy of interaction between the point mass and the *shell* is the integral of Eq. (13.20) over the whole sphere as ϕ varies from 0 to π (not 2π !) and s varies from $r - R$ to $r + R$. To carry out the integration, we have to express the integrand in terms of a single variable; we choose s . To express ϕ and $d\phi$ in terms of s , we have to do a little more geometry. Figure 13.23b shows that s is the hypotenuse of a right triangle with sides $(r - R \cos \phi)$ and $R \sin \phi$, so the Pythagorean theorem gives

$$\begin{aligned} s^2 &= (r - R \cos \phi)^2 + (R \sin \phi)^2 \\ &= r^2 - 2rR \cos \phi + R^2 \end{aligned} \quad (13.21)$$

We take differentials of both sides:

$$2s ds = 2rR \sin \phi d\phi$$

Next we divide this by $2rR$ and substitute the result into Eq. (13.20):

$$dU = -\frac{GMm}{2s} \frac{s ds}{rR} = -\frac{GMm}{2rR} ds \quad (13.22)$$

We can now integrate Eq. (13.22), recalling that s varies from $r - R$ to $r + R$:

$$U = -\frac{GMm}{2rR} \int_{r-R}^{r+R} ds = -\frac{GMm}{2rR} [(r+R) - (r-R)] \quad (13.23)$$

Finally, we have

$$U = -\frac{GMm}{r} \quad (\text{point mass } m \text{ outside spherical shell } M) \quad (13.24)$$

This is equal to the potential energy of two point masses m and M at a distance r . So we have proved that the gravitational potential energy of spherical shell M and point mass m at any distance r is the same as though they were point masses. Because the force is given by $F_r = -dU/dr$, the force is also the same.

The Gravitational Force Between Spherical Mass Distributions

Any spherically symmetric mass distribution can be thought of as a combination of concentric spherical shells. Because of the principle of superposition of forces, what is true of one shell is also true of the combination. So we have proved half of what we set out to prove: that the gravitational interaction between any spherically symmetric mass distribution and a point mass is the same as though all the mass of the spherically symmetric distribution were concentrated at its center.

The other half is to prove that *two* spherically symmetric mass distributions interact as though both were points. That's easier. In Fig. 13.23a the forces the two objects exert on each other are an action–reaction pair, and they obey Newton's third law. So we have also proved that the force that m exerts on sphere M is the same as though M were a point. But now if we replace m with a spherically symmetric mass distribution centered at m 's location, the resulting gravitational force on any part of M is the same as before, and so is the total force. This completes our proof.

A Point Mass Inside a Spherical Shell

We assumed at the beginning that the point mass m was outside the spherical shell, so our proof is valid only when m is outside a spherically symmetric mass distribution. When m is *inside* a spherical shell, the geometry is as shown in **Fig. 13.24**. The entire analysis goes just as before; Eqs. (13.18) through (13.22) are still valid. But when we get to Eq. (13.23), the limits of integration have to be changed to $R - r$ and $R + r$. We then have

$$U = -\frac{GMm}{2rR} \int_{R-r}^{R+r} ds = -\frac{GMm}{2rR} [(R+r) - (R-r)] \quad (13.25)$$

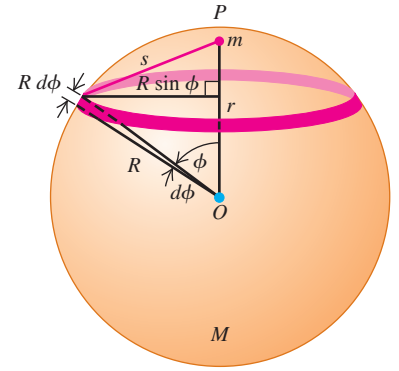
and the final result is

$$U = -\frac{GMm}{R} \quad (\text{point mass } m \text{ inside spherical shell } M) \quad (13.26)$$

Compare this result to Eq. (13.24): Instead of having r , the distance between m and the center of M , in the denominator, we have R , the radius of the shell. This means that U in Eq. (13.26) doesn't depend on r and thus has the same value everywhere inside the shell. When m moves around inside the shell, no work is done on it, so the force on m at any point inside the shell must be zero.

More generally, at any point in the interior of any spherically symmetric mass distribution (not necessarily a shell), at a distance r from its center, the gravitational force on a point mass m is the same as though we removed all the mass at points farther than r from the center and concentrated all the remaining mass at the center.

Figure 13.24 When a point mass m is *inside* a uniform spherical shell of mass M , the potential energy is the same no matter where inside the shell the point mass is located. The force from the masses' mutual gravitational interaction is zero.

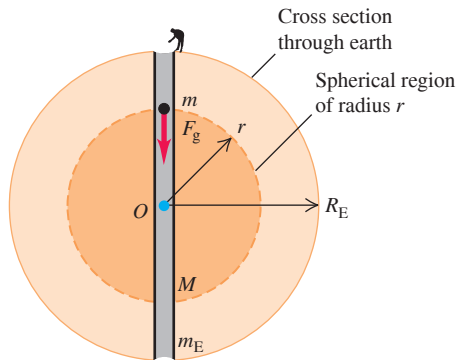


EXAMPLE 13.10 “Journey to the center of the earth”

Imagine that we drill a hole through the earth along a diameter and drop a mail pouch down the hole. Derive an expression for the gravitational force F_g on the pouch as a function of its distance from the earth's center. Assume that the earth's density is uniform (not a very realistic model; see Fig. 13.9).

IDENTIFY and SET UP From the discussion immediately above, the value of F_g at a distance r from the earth's center is determined by only the mass M within a spherical region of radius r (**Fig. 13.25**). Hence F_g

Figure 13.25 A hole through the center of the earth (assumed to be uniform). When an object is a distance r from the center, only the mass inside a sphere of radius r exerts a net gravitational force on it.



is the same as if all the mass within radius r were concentrated at the center of the earth. The mass of a uniform sphere is proportional to the volume of the sphere, which is $\frac{4}{3}\pi r^3$ for a sphere of arbitrary radius r and $\frac{4}{3}\pi R_E^3$ for the entire earth.

EXECUTE The ratio of the mass M of the sphere of radius r to the mass m_E of the earth is

$$\frac{M}{m_E} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R_E^3} = \frac{r^3}{R_E^3} \quad \text{so} \quad M = m_E \frac{r^3}{R_E^3}$$

The magnitude of the gravitational force on m is then

$$F_g = \frac{GMm}{r^2} = \frac{Gm}{r^2} \left(m_E \frac{r^3}{R_E^3} \right) = \frac{Gm_E m}{R_E^3} r$$

EVALUATE Inside this uniform-density sphere, F_g is *directly proportional* to the distance r from the center, rather than to $1/r^2$ as it is outside the sphere. At the surface $r = R_E$, we have $F_g = Gm_E m/R_E^2$, as we should. In the next chapter we'll learn how to compute the time it would take for the mail pouch to emerge on the other side of the earth.

KEYCONCEPT The gravitational force on a particle at a point a radius r from the center of a spherically symmetric mass distribution is caused solely by the mass *inside* a sphere of radius r . The mass outside radius r has no effect on the particle.

TEST YOUR UNDERSTANDING OF SECTION 13.6 In the classic 1913 science-fiction novel *At the Earth's Core*, by Edgar Rice Burroughs, explorers discover that the earth is a hollow sphere and that an entire civilization lives on the inside of the sphere. Would it be possible to stand and walk on the inner surface of a hollow, nonrotating planet?

ANSWER

No. Our analysis shows that there is *zero* gravitational force inside a hollow spherical shell. Hence visitors to the interior of a hollow planet would find themselves weightless, and they could not stand or walk on the planet's inner surface.

13.7 APPARENT WEIGHT AND THE EARTH'S ROTATION

Because the earth rotates on its axis, it is not precisely an inertial frame of reference. For this reason the apparent weight of an object on earth is not precisely equal to the earth's gravitational attraction, which we'll call the **true weight** $\vec{w}_0$ of the object. **Figure 13.26** is a cutaway view of the earth, showing three observers. Each one holds a spring scale with an object of mass m hanging from it. Each scale applies a tension force $\vec{F}$ to the object hanging from it, and the reading on each scale is the magnitude F of this force. If the observers are unaware of the earth's rotation, each one *thinks* that the scale reading equals the weight of the object because he thinks the object on his spring scale is in equilibrium. So each observer thinks that the tension $\vec{F}$ must be opposed by an equal and opposite force $\vec{w}$, which we call the **apparent weight**. But if the objects are rotating with the earth, they are *not* precisely in equilibrium. Our problem is to find the relationship between the apparent weight $\vec{w}$ and the true weight $\vec{w}_0$.

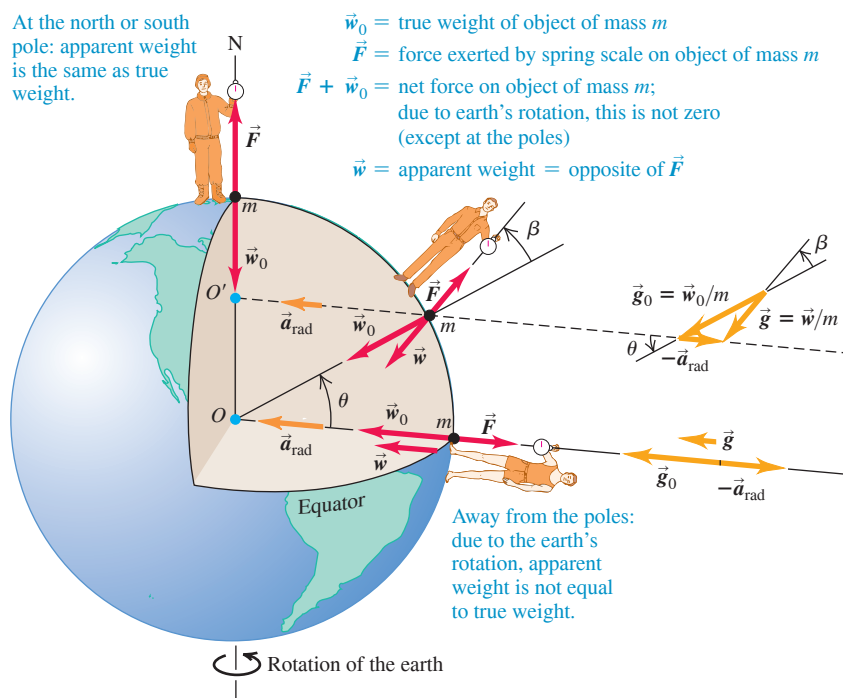
If we assume that the earth is spherically symmetric, then the true weight $\vec{w}_0$ has magnitude $Gm_E m/R_E^2$, where m_E and R_E are the mass and radius of the earth. This value is the same for all points on the earth's surface. If the center of the earth can be taken as the origin of an inertial coordinate system, then the object at the north pole really *is* in equilibrium in an inertial system, and the reading on that observer's spring scale is equal to w_0 . But the object at the equator is moving in a circle of radius R_E with speed v , and there must be a net inward force equal to the mass times the centripetal acceleration:

$$w_0 - F = \frac{mv^2}{R_E}$$

So the magnitude of the apparent weight (equal to the magnitude of F) is

$$w = w_0 - \frac{mv^2}{R_E} \quad (\text{at the equator}) \quad (13.27)$$

Figure 13.26 Except at the poles, the reading for an object being weighed on a scale (the *apparent weight*) is less than the gravitational force of attraction on the object (the *true weight*). The reason is that a net force is needed to provide a centripetal acceleration as the object rotates with the earth. For clarity, the illustration greatly exaggerates the angle β between the true and apparent weight vectors.



If the earth were not rotating, the object when released would have a free-fall acceleration $g_0 = w_0/m$. Since the earth *is* rotating, the falling object's actual acceleration relative to the observer at the equator is $g = w/m$. Dividing Eq. (13.27) by m and using these relationships, we find

$$g = g_0 - \frac{v^2}{R_E} \quad (\text{at the equator})$$

To evaluate v^2/R_E , we note that in 86,164 s a point on the equator moves a distance equal to the earth's circumference, $2\pi R_E = 2\pi(6.37 \times 10^6 \text{ m})$. (The solar day, 86,400 s, is $\frac{1}{365}$ longer than this because in one day the earth also completes $\frac{1}{365}$ of its orbit around the sun.) Thus we find

$$v = \frac{2\pi(6.37 \times 10^6 \text{ m})}{86,164 \text{ s}} = 465 \text{ m/s}$$

$$\frac{v^2}{R_E} = \frac{(465 \text{ m/s})^2}{6.37 \times 10^6 \text{ m}} = 0.0339 \text{ m/s}^2$$

So for a spherically symmetric earth the acceleration due to gravity should be about 0.03 m/s^2 less at the equator than at the poles.

At locations intermediate between the equator and the poles, the true weight $\vec{w}_0$ and the centripetal acceleration are not along the same line, and we need to write a vector equation corresponding to Eq. (13.27). From Fig. 13.26 we see that the appropriate equation is

$$\vec{w} = \vec{w}_0 - m\vec{a}_{\text{rad}} = m\vec{g}_0 - m\vec{a}_{\text{rad}} \quad (13.28)$$

The difference in the magnitudes of g and g_0 lies between zero and 0.0339 m/s^2 . As Fig. 13.26 shows, the *direction* of the apparent weight differs from the direction toward the center of the earth by a small angle β , which is 0.1° or less.

Table 13.1 gives the values of g at several locations. In addition to moderate variations with latitude, there are small variations due to elevation, differences in local density, and the earth's deviation from perfect spherical symmetry.

TABLE 13.1 Variations of g with Latitude and Elevation

Station	North Latitude	Elevation (m)	g (m/s^2)
Canal Zone	09°	0	9.78243
Jamaica	18°	0	9.78591
Bermuda	32°	0	9.79806
Denver, CO	40°	1638	9.79609
Pittsburgh, PA	40.5°	235	9.80118
Cambridge, MA	42°	0	9.80398
Greenland	70°	0	9.82534

TEST YOUR UNDERSTANDING OF SECTION 13.7 Imagine a planet that has the same mass and radius as the earth but that makes 10 rotations during the time the earth makes one rotation. What would be the difference between the acceleration due to gravity at the planet's equator and the acceleration due to gravity at its poles? (i) 0.00339 m/s^2 ; (ii) 0.0339 m/s^2 ; (iii) 0.339 m/s^2 ; (iv) 3.39 m/s^2 .

ANSWER (iv) The discussion following Eq. (13.27) shows that the difference between the acceleration due to gravity at the equator and at the poles is v^2/R_E . Since this planet has the same radius and hence the same circumference as the earth, the speed v at its equator must be 10 times the speed at the earth's equator. Hence v^2/R_E is $10^2 = 100$ times greater than for the earth, or $100(0.0339 \text{ m/s}^2) = 3.39 \text{ m/s}^2$. The acceleration due to gravity at the poles is 9.80 m/s^2 , while at the equator it is dramatically less, $9.80 \text{ m/s}^2 - 3.39 \text{ m/s}^2 = 6.41 \text{ m/s}^2$. You can show that if this planet were to rotate 17.0 times faster than the earth, the acceleration due to gravity at the equator would be zero and loose objects would fly off the equator's surface!

13.8 BLACK HOLES

In 1916 Albert Einstein presented his general theory of relativity, which included a new concept of the nature of gravitation. In his theory, a massive object actually changes the geometry of the space around it. Other objects sense this altered geometry and respond by being attracted to the first object. The general theory of relativity is beyond our scope in this chapter, but we can look at one of its most startling predictions: the existence of **black holes**,

objects whose gravitational influence is so great that nothing—not even light—can escape them. We can understand the basic idea of a black hole by using Newtonian principles.

The Escape Speed from a Star

Think first about the properties of our own sun. Its mass $M = 1.99 \times 10^{30}$ kg and radius $R = 6.96 \times 10^8$ m are much larger than those of any planet, but compared to other stars, our sun is not exceptionally massive. You can find the sun's average density ρ in the same way we found the average density of the earth in Section 13.2:

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3} = \frac{1.99 \times 10^{30} \text{ kg}}{\frac{4}{3}\pi (6.96 \times 10^8 \text{ m})^3} = 1410 \text{ kg/m}^3$$

The sun's temperatures range from 5800 K (about 5500°C or 10,000°F) at the surface up to 1.5×10^7 K (about 2.7×10^7 °F) in the interior, so it surely contains no solids or liquids. Yet gravitational attraction pulls the sun's gas atoms together until the sun is, on average, 41% denser than water and about 1200 times as dense as the air we breathe.

Now think about the escape speed for an object at the surface of the sun. In Example 13.5 (Section 13.3) we found that the escape speed from the surface of a spherical mass M with radius R is $v = \sqrt{2GM/R}$. Substituting $M = \rho V = \rho(\frac{4}{3}\pi R^3)$ into the expression for escape speed gives

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{8\pi G\rho}{3}} R \quad (13.29)$$

Using either form of this equation, you can show that the escape speed for an object at the surface of our sun is $v = 6.18 \times 10^5$ m/s (about 2.2 million km/h, or 1.4 million mi/h). This value, roughly $\frac{1}{500}$ the speed of light in vacuum, is independent of the mass of the escaping object; it depends on only the mass and radius (or average density and radius) of the sun.

Now consider various stars with the same average density ρ and different radii R . Equation (13.29) shows that for a given value of density ρ , the escape speed v is directly proportional to R . In 1783 the Rev. John Michell noted that if an object with the same average density as the sun had about 500 times the radius of the sun, its escape speed would be greater than the speed of light in vacuum, c . With his statement that “all light emitted from such a body would be made to return towards it,” Michell became the first person to suggest the existence of what we now call a black hole.

Black Holes, the Schwarzschild Radius, and the Event Horizon

The first expression for escape speed in Eq. (13.29) suggests that an object of mass M will act as a black hole if its radius R is less than or equal to a certain critical radius. How can we determine this critical radius? You might think that you can find the answer by simply setting $v = c$ in Eq. (13.29). As a matter of fact, this does give the correct result, but only because of two compensating errors. The kinetic energy of light is *not* $mc^2/2$, and the gravitational potential energy near a black hole is *not* given by Eq. (13.9). In 1916, Karl Schwarzschild used Einstein's general theory of relativity to derive an expression for the critical radius R_S , now called the **Schwarzschild radius**. The result turns out to be the same as though we had set $v = c$ in Eq. (13.29), so

$$c = \sqrt{\frac{2GM}{R_S}}$$

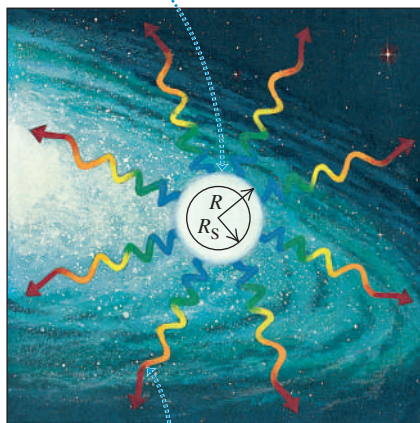
Solving for the Schwarzschild radius R_S , we find

$$\text{Schwarzschild radius of a black hole } R_S = \frac{2GM}{c^2} \quad (13.30)$$

Gravitational constant
Mass of black hole
Speed of light in vacuum

Figure 13.27 (a) An object with a radius R greater than the Schwarzschild radius R_S . (b) If the object collapses to a radius smaller than R_S , it is a black hole with an escape speed greater than the speed of light. The surface of the sphere of radius R_S is called the event horizon of the black hole.

(a) When the radius R of an object is greater than the Schwarzschild radius R_S , light can escape from the surface of the object.



Gravity acting on the escaping light “red shifts” it to longer wavelengths.

(b) If all the mass of the object lies inside radius R_S , the object is a black hole: No light can escape from it.



If a spherical, nonrotating object with mass M has a radius less than R_S , then *nothing* (not even light) can escape from the surface of the object, and the object is a black hole (Fig. 13.27). In this case, any other object within a distance R_S of the center of the black hole is trapped by the gravitational attraction of the black hole and cannot escape from it.

The surface of the sphere with radius R_S surrounding a black hole is called the **event horizon**: Since light can't escape from within that sphere, we can't see events occurring inside. All that an observer outside the event horizon can know about a black hole is its mass (from its gravitational effects on other objects), its electric charge (from the electric forces it exerts on other charged objects), and its angular momentum (because a rotating black hole tends to drag space—and everything in that space—around with it). All other information about the object is irretrievably lost when it collapses inside its event horizon.

EXAMPLE 13.11 Black hole calculations

Astrophysical theory suggests that a burned-out star whose mass is at least three solar masses will collapse under its own gravity to form a black hole. If it does, what is the radius of its event horizon?

IDENTIFY, SET UP, and EXECUTE The radius in question is the Schwarzschild radius. We use Eq. (13.30) with a value of M equal to three solar masses, or $M = 3(1.99 \times 10^{30} \text{ kg}) = 6.0 \times 10^{30} \text{ kg}$:

$$\begin{aligned} R_S &= \frac{2GM}{c^2} = \frac{2(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(6.0 \times 10^{30} \text{ kg})}{(3.00 \times 10^8 \text{ m/s})^2} \\ &= 8.9 \times 10^3 \text{ m} = 8.9 \text{ km} = 5.5 \text{ mi} \end{aligned}$$

EVALUATE The average density of such an object is

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{6.0 \times 10^{30} \text{ kg}}{\frac{4}{3}\pi (8.9 \times 10^3 \text{ m})^3} = 2.0 \times 10^{18} \text{ kg/m}^3$$

This is about 10^{15} times as great as the density of familiar matter on earth and is comparable to the densities of atomic nuclei. In fact, once the object collapses to a radius of R_S , nothing can prevent it from collapsing further. All of the mass ends up being crushed down to a single point called a *singularity* at the center of the event horizon. This point has zero volume and so has *infinite* density.

KEYCONCEPT The event horizon of a black hole has a radius of R_S , the Schwarzschild radius, which is proportional to the mass of the black hole. Nothing can escape from inside the event horizon.

Figure 13.28 In a *binary* star system, two stars orbit each other; in the special case shown here, one of the stars is a black hole. The black hole itself cannot be seen, but the x rays from its accretion disk can be detected.

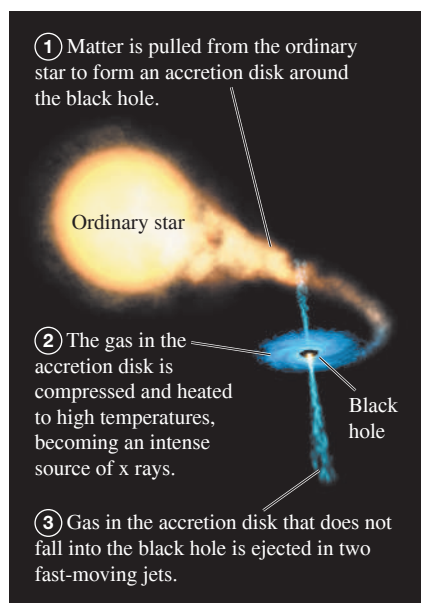
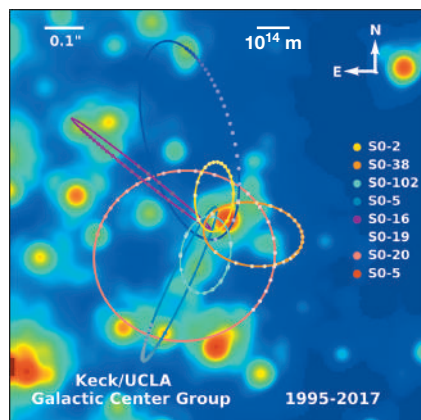


Figure 13.29 This false-color image shows the motions of stars at the center of our galaxy over a 17-year period. Analysis of these orbits by using Kepler's third law indicates that the stars are moving about an unseen object that is some 4.1×10^6 times the mass of the sun. The scale bar indicates a length of 10^{14} m (670 times the distance from the earth to the sun).



A Visit to a Black Hole

At points far from a black hole, its gravitational effects are the same as those of any normal object with the same mass. If the sun collapsed to form a black hole, the orbits of the planets would be unaffected. But things get dramatically different close to the black hole. If you decided to become a martyr for science and jump into a black hole, the friends you left behind would notice several odd effects as you moved toward the event horizon, most of them associated with effects of general relativity.

If you carried a radio transmitter to send back your comments on what was happening, your friends would have to retune their receiver continuously to lower and lower frequencies, an effect called the *gravitational red shift*. Consistent with this shift, they would observe that your clocks (electronic or biological) would appear to run more and more slowly, an effect called *time dilation*. In fact, during their lifetimes they would never see you make it to the event horizon.

In your frame of reference, you would make it to the event horizon in a rather short time but in a rather disquieting way. As you fell feet first into the black hole, the gravitational pull on your feet would be greater than that on your head, which would be slightly farther away from the black hole. The *differences* in gravitational force on different parts of your body would be great enough to stretch you along the direction toward the black hole and compress you perpendicular to it. These effects (called *tidal forces*) would rip you to atoms, and then rip your atoms apart, before you reached the event horizon.

Detecting Black Holes

If light cannot escape from a black hole and if black holes are as small as Example 13.11 suggests, how can we know that such things exist? The answer is that any gas or dust near the black hole tends to be pulled into an *accretion disk* that swirls around and into the black hole, rather like a whirlpool (**Fig. 13.28**). Friction within the accretion disk's gas causes it to lose mechanical energy and spiral into the black hole; as it moves inward, it is compressed together. This causes heating of the gas, just as air compressed in a bicycle pump gets hotter. Temperatures in excess of 10^6 K can occur in the accretion disk, so hot that the disk emits not just visible light (as do objects that are “red-hot” or “white-hot”) but x rays. Astronomers look for these x rays (emitted by the gas material *before* it crosses the event horizon) to signal the presence of a black hole. Several promising candidates have been found, and astronomers now express considerable confidence in the existence of black holes.

Black holes in binary star systems like the one depicted in Fig. 13.28 have masses a few times greater than the sun's mass. There is also mounting evidence for the existence of much larger *supermassive black holes*. One example lies at the center of our Milky Way galaxy, some 26,000 light-years from earth in the direction of the constellation Sagittarius. High-resolution images of the galactic center reveal stars moving at speeds greater than 1500 km/s about an unseen object that lies at the position of a source of radio waves called Sgr A* (**Fig. 13.29**). By analyzing these motions, astronomers can infer the period T and semi-major axis a of each star's orbit. The mass m_X of the unseen object can be calculated from Kepler's third law in the form given in Eq. (13.17), with the mass of the sun m_S replaced by m_X :

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_X}} \quad \text{so} \quad m_X = \frac{4\pi^2 a^3}{GT^2}$$

The conclusion is that the mysterious dark object at the galactic center has a mass of 8.2×10^36 kg, or 4.1 *million* times the mass of the sun. Yet observations with radio telescopes show that it has a radius no more than 4.4×10^{10} m, about one-third of the distance from the earth to the sun. These observations suggest that this massive, compact object is a black hole with a Schwarzschild radius of 1.1×10^{10} m. Astronomers hope to improve the resolution of their observations so that they can actually see the event horizon of this black hole.

Additional evidence for the existence of black holes has come from observations of *gravitational radiation*. Einstein's general theory of relativity, which we'll discuss in Chapter 37, predicts that space itself is curved by the presence of massive objects like a planet, star, or black hole. If a massive object accelerates in a certain manner, it produces ripples in the curvature of space that radiate outward from the object. The disturbances caused by such gravitational radiation are incredibly feeble, but can be measured with sensitive detectors if the objects that produce them are very massive and have tremendous accelerations. This can happen when two massive black holes are in close orbits around each other. Each black hole has an acceleration as it moves around its curved orbit, so it emits gravitational radiation that carries away energy. This makes the orbits of the black holes smaller, so they move faster and emit gravitational radiation at an ever faster rate. The orbits keep shrinking until the two black holes finally merge. Since 2015 scientists have detected the gravitational radiation from several such black hole mergers. From the data, they conclude that the merging black holes have masses from 7 to 36 times the mass of the sun. (Rainer Weiss, Kip Thorne, and Barry Barish were awarded the 2017 Nobel Prize in Physics for their contributions to these discoveries.)

Other lines of research suggest that even larger black holes, in excess of 10^9 times the mass of the sun, lie at the centers of other galaxies. Observational and theoretical studies of black holes of all sizes continue to be an exciting area of research in both physics and astronomy.

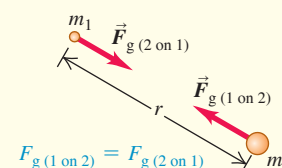
TEST YOUR UNDERSTANDING OF SECTION 13.8 If the sun somehow collapsed to form a black hole, what effect would this event have on the orbit of the earth? The orbit would (i) shrink; (ii) expand; (iii) remain the same size.

ANSWER (iii) If the sun collapsed into a black hole (which, according to our understanding of stars, it can-
not do), the sun would have a much smaller radius but the same mass. The sun's gravitational force
on the earth doesn't depend on the sun's radius, so the earth's orbit would be unaffected.

CHAPTER 13 SUMMARY

Newton's law of gravitation: Any two particles with masses m_1 and m_2 , a distance r apart, attract each other with forces inversely proportional to r^2 . These forces form an action–reaction pair and obey Newton's third law. When two or more objects exert gravitational forces on a particular object, the total gravitational force on that individual object is the vector sum of the forces exerted by the other objects. The gravitational interaction between spherical mass distributions, such as planets or stars, is the same as if all the mass of each distribution were concentrated at the center. (See Examples 13.1–13.3 and 13.10.)

$$F_g = \frac{Gm_1m_2}{r^2} \quad (13.1)$$



Gravitational force, weight, and gravitational potential energy: The weight w of an object is the total gravitational force exerted on it by all other objects in the universe. Near the surface of the earth (mass m_E and radius R_E), the weight is essentially equal to the gravitational force of the earth alone. The gravitational potential energy U of two masses m and m_E separated by a distance r is inversely proportional to r . The potential energy is never positive; it is zero only when the two objects are infinitely far apart. (See Examples 13.4 and 13.5.)

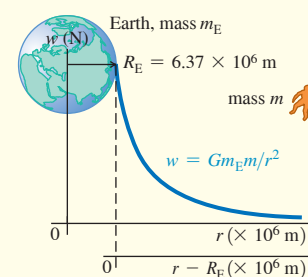
$$w = F_g = \frac{Gm_E m}{R_E^2} \quad (13.3)$$

(weight at earth's surface)

$$g = \frac{Gm_E}{R_E^2} \quad (13.4)$$

(acceleration due to gravity at earth's surface)

$$U = -\frac{Gm_E m}{r} \quad (13.9)$$



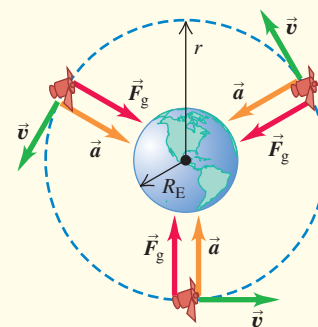
Orbits: When a satellite moves in a circular orbit, the centripetal acceleration is provided by the gravitational attraction of the earth. Kepler's three laws describe the more general case: an elliptical orbit of a planet around the sun or a satellite around a planet. (See Examples 13.6–13.9.)

$$v = \sqrt{\frac{Gm_E}{r}} \quad (13.10)$$

(speed in circular orbit)

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{Gm_E}} = \frac{2\pi r^{3/2}}{\sqrt{Gm_E}} \quad (13.12)$$

(period in circular orbit)



Black holes: If a nonrotating spherical mass distribution with total mass M has a radius less than its Schwarzschild radius R_S , it is called a black hole. The gravitational interaction prevents anything, including light, from escaping from within a sphere with radius R_S . (See Example 13.11.)

$$R_S = \frac{2GM}{c^2} \quad (13.30)$$

(Schwarzschild radius)



If all of the object is inside its Schwarzschild radius $R_S = 2GM/c^2$, the object is a black hole.

Chapter 13 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 13.5** (Section 13.3) before attempting these problems.

VP13.5.1 The *New Horizons* spacecraft (mass 478 kg) was launched from earth in 2006 and flew past Pluto in 2015. What minimum amount of work has to be done on a spacecraft of this mass to send it from the earth's surface to infinitely far away from the earth? Neglect air resistance and the gravitational effects of the sun, moon, and other planets.

VP13.5.2 A piece of spacecraft debris initially at rest falls to the earth's surface from a height above the earth equal to one-half of the earth's radius. Find the speed at which the piece of debris hits the surface. Neglect air resistance and the gravitational pull of the moon.

VP13.5.3 An astronaut on Mars (mass 6.42×10^{23} kg, radius 3.39×10^6 m) launches a probe straight upward from the surface at 3.00×10^3 m/s. What is the maximum height above the surface that the probe reaches? Neglect air resistance and the gravitational pull of the two small moons of Mars.

VP13.5.4 In order to rendezvous with an asteroid passing close to the earth, a spacecraft must be moving at 8.50×10^3 m/s relative to the earth at a distance of 2.50×10^8 m from the center of the earth. At what speed must this spacecraft be launched from the earth's surface? Neglect air resistance and the gravitational pull of the moon.

Be sure to review **EXAMPLE 13.6** (Section 13.4) before attempting these problems.

VP13.6.1 You wish to place a spacecraft in a circular orbit around the earth so that its orbital speed will be 4.00×10^3 m/s. What are this orbit's (a) radius, (b) altitude above the earth's surface, and (c) period (in hours)?

VP13.6.2 You are designing a spacecraft intended to monitor a human expedition to Mars (mass 6.42×10^{23} kg, radius 3.39×10^6 m). This

spacecraft will orbit around the Martian equator with an orbital period of 24.66 h, the same as the rotation period of Mars, so that it will always be above the same point on the equator. (a) What must be the radius of the orbit? (b) What will be the speed of the spacecraft in its orbit?

VP13.6.3 A spacecraft of mass 1.00×10^3 kg orbits the sun (mass 1.99×10^{30} kg) in a circular orbit of radius 1.50×10^{11} m (equal to the average distance from the sun to the earth). You wish to move the spacecraft into a smaller circular orbit around the sun of radius 1.08×10^{11} m (equal to the average distance from the sun to Venus). In doing this, what will be the changes in (a) the spacecraft's kinetic energy, (b) the gravitational potential energy of the sun–spacecraft system, and (c) the total mechanical energy of the sun–spacecraft system? Neglect the gravitational pulls of the planets on the spacecraft.

VP13.6.4 A satellite of mass 1.50×10^3 kg is at point P , at an altitude of 3.50×10^6 m above the earth's surface, and traveling at 7.50×10^3 m/s. (a) If you wanted to put the satellite into a circular orbit at this altitude above the earth's surface, how much work would you have to do on it at point P ? (b) If instead you wanted to make the satellite escape the earth, how much work would you have to do on it at point P ?

Be sure to review **EXAMPLES 13.7, 13.8, and 13.9** (Section 13.5) before attempting these problems.

VP13.9.1 Some comets are in highly elongated orbits that come very close to the sun at perihelion. The distance from one such comet to the center of the sun is 6.00×10^9 m at perihelion and 3.00×10^{12} m at aphelion. For this comet's orbit, find (a) the semi-major axis, (b) the eccentricity, and (c) the period (in years).

VP13.9.2 The orbit of a certain asteroid around the sun has period 7.85 y and eccentricity 0.250. Find (a) the semi-major axis and the distances from the sun to the asteroid at (b) perihelion and (c) aphelion.

VP13.9.3 The minor planet (33340) 1998 VG₄₄ makes exactly two orbits of the sun during the time that Neptune makes three orbits. The eccentricity of the orbit of this minor planet is 0.25, and the orbit radius of Neptune is 4.5×10^{12} m. (a) Find the semi-major axis of this minor planet. (b) Find the distance between the sun and the minor planet at perihelion. At perihelion, is this minor planet inside or outside the orbit of Neptune?

VP13.9.4 In 2017 astronomers discovered a planet orbiting the star HATS-43. The orbit of the planet around HATS-43 has semi-major axis 7.41×10^9 m, eccentricity 0.173, and period 4.39 days. Find (a) the distance between HATS-43 and the planet at its closest approach and (b) the mass of HATS-43. (For comparison, the distance between the sun and Mercury at its closest approach is 4.60×10^{10} m and the mass of the sun is 1.99×10^{30} kg.)

BRIDGING PROBLEM Speeds in an Elliptical Orbit

A comet orbits the sun (mass m_S) in an elliptical orbit of semi-major axis a and eccentricity e . (a) Find expressions for the speeds of the comet at perihelion and aphelion. (b) Evaluate these expressions for Comet Halley (see Example 13.9), and find the kinetic energy, gravitational potential energy, and total mechanical energy for this comet at perihelion and aphelion. Take the mass of Comet Halley to be 2.2×10^{14} kg.

SOLUTION GUIDE

IDENTIFY and SET UP

1. Sketch the situation; show all relevant dimensions. Label the perihelion and aphelion. (See Figure 13.18.)
2. List the unknown quantities, and identify the target variables.
3. Just as for a satellite orbiting the earth, the total mechanical energy is conserved for a comet orbiting the sun. (Why?) What other quantity is conserved as the comet moves in its orbit? (*Hint*: See Section 13.5.)

EXECUTE

4. You'll need at least two equations that involve the two unknown speeds, and you'll need expressions for the sun–comet distances at perihelion and aphelion. (*Hint*: See Fig. 13.18.)
5. Solve the equations for your target variables. Compare your expressions: Which speed is lower? Does this make sense?
6. Use your expressions from step 5 to find the perihelion and aphelion speeds for Comet Halley. (*Hint*: See Appendix F.)
7. Use your results from step 6 to find the kinetic energy K , gravitational potential energy U , and total mechanical energy E for Comet Halley at perihelion and aphelion.

EVALUATE

8. Check whether your results from part (a) make sense for the special case of a circular orbit ($e = 0$).
9. In part (b), how do your calculated values of E at perihelion and aphelion compare? Does this make sense? What does it mean that E is negative?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q13.1 A student wrote: “The only reason an apple falls downward to meet the earth instead of the earth rising upward to meet the apple is that the earth is much more massive and so exerts a much greater pull.” Please comment.

Q13.2 If all planets had the same average density, how would the acceleration due to gravity at the surface of a planet depend on its radius?

Q13.3 Is a pound of butter on the earth the same amount as a pound of butter on Mars? What about a kilogram of butter? Explain.

Q13.4 Example 13.2 (Section 13.1) shows that the acceleration of each sphere caused by the gravitational force is inversely proportional to the mass of that sphere. So why does the force of gravity give all masses the same acceleration when they are dropped near the surface of the earth?

Q13.5 When will you attract the sun more: today at noon, or tonight at midnight? Explain.

Q13.6 Since the moon is constantly attracted toward the earth by the gravitational interaction, why doesn't it crash into the earth?

Q13.7 A spaceship makes a circular orbit with period T around a star. If it were to orbit, at the same distance, a star with three times the mass of the original star, would the new period (in terms of T) be (a) $3T$, (b) $T\sqrt{3}$, (c) T , (d) $T/\sqrt{3}$, or (e) $T/3$?

Q13.8 A planet makes a circular orbit with period T around a star. If the planet were to orbit at the same distance around this star, but the planet had three times as much mass, what would the new period (in terms of T) be: (a) $3T$, (b) $T\sqrt{3}$, (c) T , (d) $T/\sqrt{3}$, or (e) $T/3$?

Q13.9 The sun pulls on the moon with a force that is more than twice the magnitude of the force with which the earth attracts the moon. Why, then, doesn't the sun take the moon away from the earth?

Q13.10 Which takes more fuel: a voyage from the earth to the moon or from the moon to the earth? Explain.

Q13.11 A planet is moving at constant speed in a circular orbit around a star. In one complete orbit, what is the net amount of work done on the planet by the star's gravitational force: positive, negative, or zero? What if the planet's orbit is an ellipse, so that the speed is not constant? Explain your answers.

Q13.12 Does the escape speed for an object at the earth's surface depend on the direction in which it is launched? Explain. Does your answer depend on whether or not you include the effects of air resistance?

Q13.13 If a projectile is fired straight up from the earth's surface, what would happen if the total mechanical energy (kinetic plus potential) is (a) less than zero, and (b) greater than zero? In each case, ignore air resistance and the gravitational effects of the sun, the moon, and the other planets.

Q13.14 Discuss whether this statement is correct: “In the absence of air resistance, the trajectory of a projectile thrown near the earth’s surface is an *ellipse*, not a parabola.”

Q13.15 The earth is closer to the sun in November than in May. In which of these months does it move faster in its orbit? Explain why.

Q13.16 A communications firm wants to place a satellite in orbit so that it is always directly above the earth’s 45th parallel (latitude 45° north). This means that the plane of the orbit will not pass through the center of the earth. Is such an orbit possible? Why or why not?

Q13.17 At what point in an elliptical orbit is the acceleration maximum? At what point is it minimum? Justify your answers.

Q13.18 What would Kepler’s third law be for circular orbits if an amendment to Newton’s law of gravitation made the gravitational force inversely proportional to r^3 ? Would this change affect Kepler’s other two laws? Explain.

Q13.19 In the elliptical orbit of Comet Halley shown in Fig. 13.21a, the sun’s gravity is responsible for making the comet fall inward from aphelion to perihelion. But what is responsible for making the comet move from perihelion back outward to aphelion?

Q13.20 Many people believe that orbiting astronauts feel weightless because they are “beyond the pull of the earth’s gravity.” How far from the earth would a spacecraft have to travel to be truly beyond the earth’s gravitational influence? If a spacecraft were really unaffected by the earth’s gravity, would it remain in orbit? Explain. What is the real reason astronauts in orbit feel weightless?

Q13.21 As part of their training before going into orbit, astronauts ride in an airliner that is flown along the same parabolic trajectory as a freely falling projectile. Explain why this gives the same experience of apparent weightlessness as being in orbit.

EXERCISES

Section 13.1 Newton’s Law of Gravitation

13.1 • What is the ratio of the gravitational pull of the sun on the moon to that of the earth on the moon? (Assume the distance of the moon from the sun can be approximated by the distance of the earth from the sun.) Use the data in Appendix F. Is it more accurate to say that the moon orbits the earth, or that the moon orbits the sun?

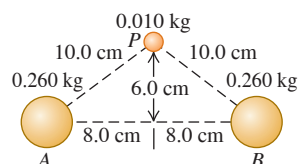
13.2 • You are sitting in the front row in your physics class. Estimate the gravitational force that the instructor exerts on you. Identify the assumptions and approximations you made to reach your answer. How does the magnitude that you estimate for this force compare with the gravity force exerted on you by the earth?

13.3 • **Rendezvous in Space!** A couple of astronauts agree to rendezvous in space after hours. Their plan is to let gravity bring them together. One of them has a mass of 65 kg and the other a mass of 72 kg, and they start from rest 20.0 m apart. (a) Make a free-body diagram of each astronaut, and use it to find his or her initial acceleration. As a rough approximation, we can model the astronauts as uniform spheres. (b) If the astronauts’ acceleration remained constant, how many days would they have to wait before reaching each other? (Careful! They *both* have acceleration toward each other.) (c) Would their acceleration, in fact, remain constant? If not, would it increase or decrease? Why?

13.4 • Two uniform spheres, each with mass M and radius R , touch each other. What is the magnitude of their gravitational force of attraction?

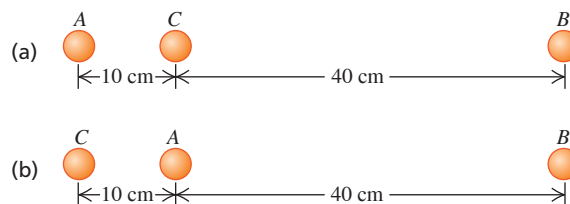
13.5 • Two uniform spheres, each of mass 0.260 kg, are fixed at points A and B (Fig. E13.5). Find the magnitude and direction of the initial acceleration of a uniform sphere with mass 0.010 kg if released from rest at point P and acted on only by forces of gravitational attraction of the spheres at A and B.

Figure E13.5



13.6 • Find the magnitude and direction of the net gravitational force on mass A due to masses B and C in Fig. E13.6. Each mass is 2.00 kg.

Figure E13.6



13.7 • A typical adult human has a mass of about 70 kg. (a) What force does a full moon exert on such a human when it is directly overhead with its center 378,000 km away? (b) Compare this force with the force exerted on the human by the earth.

13.8 • An 8.00 kg point mass and a 12.0 kg point mass are held in place 50.0 cm apart. A particle of mass m is released from a point between the two masses 20.0 cm from the 8.00 kg mass along the line connecting the two fixed masses. Find the magnitude and direction of the acceleration of the particle.

13.9 • A particle of mass $3m$ is located 1.00 m from a particle of mass m . (a) Where should you put a third mass M so that the net gravitational force on M due to the two masses is exactly zero? (b) Is the equilibrium of M at this point stable or unstable (i) for points along the line connecting m and $3m$, and (ii) for points along the line passing through M and perpendicular to the line connecting m and $3m$?

13.10 • The point masses m and $2m$ lie along the x -axis, with m at the origin and $2m$ at $x = L$. A third point mass M is moved along the x -axis. (a) At what point is the net gravitational force on M due to the other two masses equal to zero? (b) Sketch the x -component of the net force on M due to m and $2m$, taking quantities to the right as positive. Include the regions $x < 0$, $0 < x < L$, and $x > L$. Be especially careful to show the behavior of the graph on either side of $x = 0$ and $x = L$.

Section 13.2 Weight

13.11 • At what distance above the surface of the earth is the acceleration due to the earth’s gravity 0.980 m/s^2 if the acceleration due to gravity at the surface has magnitude 9.80 m/s^2 ?

13.12 • The mass of Venus is 81.5% that of the earth, and its radius is 94.9% that of the earth. (a) Compute the acceleration due to gravity on the surface of Venus from these data. (b) If a rock weighs 75.0 N on earth, what would it weigh at the surface of Venus?

13.13 • Titania, the largest moon of the planet Uranus, has $\frac{1}{8}$ the radius of the earth and $\frac{1}{1700}$ the mass of the earth. (a) What is the acceleration due to gravity at the surface of Titania? (b) What is the average density of Titania? (This is less than the density of rock, which is one piece of evidence that Titania is made primarily of ice.)

13.14 • Rhea, one of Saturn’s moons, has a radius of 764 km and an acceleration due to gravity of 0.265 m/s^2 at its surface. Calculate its mass and average density.

13.15 • A science-fiction author asks for your help. He wants to write about a newly discovered spherically symmetric planet that has the same average density as the earth but with a 25% larger radius. (a) What is g on this planet? (b) If he decides to have his explorers weigh the same on this planet as on earth, how should he change its average density?

Section 13.3 Gravitational Potential Energy

13.16 • **Volcanoes on Io.** Jupiter’s moon Io has active volcanoes (in fact, it is the most volcanically active object in the solar system) that eject material as high as 500 km (or even higher) above the surface. Io has a mass of $8.93 \times 10^{22} \text{ kg}$ and a radius of 1821 km. For this calculation, ignore any variation in gravity over the 500 km range of the debris. How high would this material go on earth if it were ejected with the same speed as on Io?

13.17 • Use the results of Example 13.5 (Section 13.3) to calculate the escape speed for a spacecraft (a) from the surface of Mars and (b) from the surface of Jupiter. Use the data in Appendix F. (c) Why is the escape speed for a spacecraft independent of the spacecraft's mass?

13.18 •• Ten days after it was launched toward Mars in December 1998, the *Mars Climate Orbiter* spacecraft (mass 629 kg) was 2.87×10^6 km from the earth and traveling at 1.20×10^4 km/h relative to the earth. At this time, what were (a) the spacecraft's kinetic energy relative to the earth and (b) the potential energy of the earth-spacecraft system?

13.19 •• A planet orbiting a distant star has radius 3.24×10^6 m. The escape speed for an object launched from this planet's surface is 7.65×10^3 m/s. What is the acceleration due to gravity at the surface of the planet?

13.20 •• Estimate the cruising altitude of an airplane during a transatlantic flight. If you are a passenger in the plane when it is at this altitude, what is the percentage change in the gravitational energy of the system of you and the earth compared to when you were in the airport waiting to board the plane? For which of your two locations is the gravitational potential energy greater? (*Hint:* Use the power series for $(1 + x)^n$ given in Appendix B.)

13.21 • You are standing on the surface of a planet that has spherical symmetry and a radius of 5.00×10^6 m. The gravitational potential energy U of the system composed of you and the planet is -1.20×10^9 J if we choose U to be zero when you are very far from the planet. What is the magnitude of the gravity force that the planet exerts on you when you are standing on its surface?

Section 13.4 The Motion of Satellites

13.22 • A satellite of mass m is in a circular orbit around a spherical planet of mass m_p . The kinetic energy of the satellite is K_A when its orbit radius is r_A . In terms of r_A , what must the orbit radius be in order for the kinetic energy of the satellite to be $2K_A$?

13.23 • Let the gravitational potential energy of the earth-satellite system be zero in the limit that the orbit radius is very large, so the total mechanical energy of the satellite-earth system is given by Eq. (13.13). If the kinetic energy of the satellite is 2.00×10^6 J, what are the total energy and the gravitational potential energy of the satellite-earth system?

13.24 • An earth satellite moves in a circular orbit with an orbital speed of 6200 m/s. Find (a) the time of one revolution of the satellite; (b) the radial acceleration of the satellite in its orbit.

13.25 • For a satellite to be in a circular orbit 890 km above the surface of the earth, (a) what orbital speed must it be given, and (b) what is the period of the orbit (in hours)?

13.26 •• Aura Mission. On July 15, 2004 NASA launched the *Aura* spacecraft to study the earth's climate and atmosphere. This satellite was injected into an orbit 705 km above the earth's surface. Assume a circular orbit. (a) How many hours does it take this satellite to make one orbit? (b) How fast (in km/s) is the *Aura* spacecraft moving?

13.27 •• Two satellites are in circular orbits around a planet that has radius 9.00×10^6 m. One satellite has mass 68.0 kg, orbital radius 7.00×10^7 m, and orbital speed 4800 m/s. The second satellite has mass 84.0 kg and orbital radius 3.00×10^7 m. What is the orbital speed of this second satellite?

13.28 •• International Space Station. In its orbit each day, the International Space Station makes 15.65 revolutions around the earth. Assuming a circular orbit, how high is this satellite above the surface of the earth?

13.29 • Deimos, a moon of Mars, is about 12 km in diameter with mass 1.5×10^{15} kg. Suppose you are stranded alone on Deimos and want to play a one-person game of baseball. You would be the pitcher, and you would be the batter! (a) With what speed would you have to throw a baseball so that it would go into a circular orbit just above the surface and return to you so you could hit it? Do you think you could actually throw it at this speed? (b) How long (in hours) after throwing the ball should you be ready to hit it? Would this be an action-packed baseball game?

Section 13.5 Kepler's Laws and the Motion of Planets

13.30 •• Hot Jupiters. In 2004 astronomers reported the discovery of a large Jupiter-sized planet orbiting very close to the star HD 179949 (hence the term "hot Jupiter"). The orbit was just $\frac{1}{9}$ the distance of Mercury from our sun, and it takes the planet only 3.09 days to make one orbit (assumed to be circular). (a) What is the mass of the star? Express your answer in kilograms and as a multiple of our sun's mass. (b) How fast (in km/s) is this planet moving?

13.31 •• The star Rho¹ Cancri is 57 light-years from the earth and has a mass 0.85 times that of our sun. A planet has been detected in a circular orbit around Rho¹ Cancri with an orbital radius equal to 0.11 times the radius of the earth's orbit around the sun. What are (a) the orbital speed and (b) the orbital period of the planet of Rho¹ Cancri?

13.32 •• In March 2006, two small satellites were discovered orbiting Pluto, one at a distance of 48,000 km and the other at 64,000 km. Pluto already was known to have a large satellite Charon, orbiting at 19,600 km with an orbital period of 6.39 days. Assuming that the satellites do not affect each other, find the orbital periods of the two small satellites *without* using the mass of Pluto.

13.33 • The dwarf planet Pluto has an elliptical orbit with a semi-major axis of 5.91×10^{12} m and eccentricity 0.249. (a) Calculate Pluto's orbital period. Express your answer in seconds and in earth years. (b) During Pluto's orbit around the sun, what are its closest and farthest distances from the sun?

13.34 •• Planets Beyond the Solar System. On October 15, 2001, a planet was discovered orbiting around the star HD 68988. Its orbital distance was measured to be 10.5 million kilometers from the center of the star, and its orbital period was estimated at 6.3 days. What is the mass of HD 68988? Express your answer in kilograms and in terms of our sun's mass. (Consult Appendix F.)

Section 13.6 Spherical Mass Distributions

13.35 •• A thin spherical shell has radius $r_A = 4.00$ m and mass $m_A = 20.0$ kg. It is concentric with a second thin spherical shell that has radius $r_B = 6.00$ m and mass $m_B = 40.0$ kg. What is the net gravitational force that the two shells exert on a point mass of 0.0200 kg that is a distance r from the common center of the two shells, for (a) $r = 2.00$ m (inside both shells), (b) $r = 5.00$ m (in the space between the two shells), and (c) $r = 8.00$ m (outside both shells)?

13.36 • A uniform, spherical, 1000.0 kg shell has a radius of 5.00 m. (a) Find the gravitational force this shell exerts on a 2.00 kg point mass placed at the following distances from the center of the shell: (i) 5.01 m, (ii) 4.99 m, (iii) 2.72 m. (b) Sketch a qualitative graph of the magnitude of the gravitational force this sphere exerts on a point mass m as a function of the distance r of m from the center of the sphere. Include the region from $r = 0$ to $r \rightarrow \infty$.

13.37 •• A uniform, solid, 1000.0 kg sphere has a radius of 5.00 m. (a) Find the gravitational force this sphere exerts on a 2.00 kg point mass placed at the following distances from the center of the sphere: (i) 5.01 m, (ii) 2.50 m. (b) Sketch a qualitative graph of the magnitude of the gravitational force this sphere exerts on a point mass m as a function of the distance r of m from the center of the sphere. Include the region from $r = 0$ to $r \rightarrow \infty$.

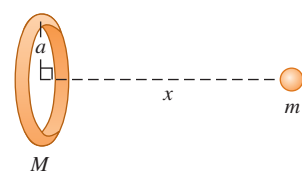
13.38 • CALC A thin, uniform rod has length L and mass M . A small uniform sphere of mass m is placed a distance x from one end of the rod, along the axis of the rod (**Fig. E13.38**). (a) Calculate the gravitational potential energy of the rod–sphere system. Take the potential energy to be zero when the rod and sphere are infinitely far apart. Show that your answer reduces to the expected result when x is much larger than L . (*Hint:* Use the power series expansion for $\ln(1 + x)$ given in Appendix B.) (b) Use $F_x = -dU/dx$ to find the magnitude and direction of the gravitational force exerted on the sphere by the rod (see Section 7.4). Show that your answer reduces to the expected result when x is much larger than L .

Figure E13.38



13.39 • CALC Consider the ring-shaped object in **Fig. E13.39**. A particle with mass m is placed a distance x from the center of the ring, along the line through the center of the ring and perpendicular to its plane. (a) Calculate the gravitational potential energy U of this system. Take the potential energy to be zero when the two objects are far apart. (b) Show that your answer to part (a) reduces to the expected result when x is much larger than the radius a of the ring. (c) Use $F_x = -dU/dx$ to find the magnitude and direction of the force on the particle (see Section 7.4). (d) Show that your answer to part (c) reduces to the expected result when x is much larger than a . (e) What are the values of U and F_x when $x = 0$? Explain why these results make sense.

Figure E13.39



13.40 • Define the gravitational field $\vec{g}$ at some point to be equal to the gravitational force $\vec{F}_g$ on a small object placed at that point divided by the mass m of the object, so $\vec{g} = \vec{F}_g/m$. A spherical shell has mass M and radius R . What is the magnitude of the gravitational field at the following distances from the center of the shell: (a) $r < R$ and (b) $r > R$?

Section 13.7 Apparent Weight and the Earth's Rotation

13.41 • The acceleration due to gravity at the north pole of Neptune is approximately 11.2 m/s^2 . Neptune has mass $1.02 \times 10^{26} \text{ kg}$ and radius $2.46 \times 10^4 \text{ km}$ and rotates once around its axis in about 16 h. (a) What is the gravitational force on a 3.00 kg object at the north pole of Neptune? (b) What is the apparent weight of this same object at Neptune's equator? (Note that Neptune's "surface" is gaseous, not solid, so it is impossible to stand on it.)

13.42 •• A Visit to Santa. You decide to visit Santa Claus at the north pole to put in a good word about your splendid behavior throughout the year. While there, you notice that the elf Sneezy, when hanging from a rope, produces a tension of 395.0 N in the rope. If Sneezy hangs from a similar rope while delivering presents at the earth's equator, what will the tension in it be? (Recall that the earth is rotating about an axis through its north and south poles.) Consult Appendix F and start with a free-body diagram of Sneezy at the equator.

Section 13.8 Black Holes

13.43 •• At the Galaxy's Core. Astronomers have observed a small, massive object at the center of our Milky Way galaxy (see Section 13.8). A ring of material orbits this massive object; the ring has a diameter of about 15 light-years and an orbital speed of about 200 km/s .

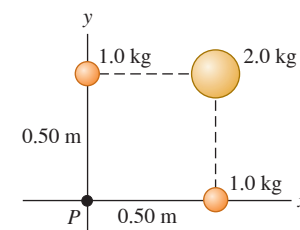
(a) Determine the mass of the object at the center of the Milky Way galaxy. Give your answer both in kilograms and in solar masses (one solar mass is the mass of the sun). (b) Observations of stars, as well as theories of the structure of stars, suggest that it is impossible for a single star to have a mass of more than about 50 solar masses. Can this massive object be a single, ordinary star? (c) Many astronomers believe that the massive object at the center of the Milky Way galaxy is a black hole. If so, what must the Schwarzschild radius of this black hole be? Would a black hole of this size fit inside the earth's orbit around the sun?

13.44 • In 2005 astronomers announced the discovery of a large black hole in the galaxy Markarian 766 having clumps of matter orbiting around once every 27 hours and moving at $30,000 \text{ km/s}$. (a) How far are these clumps from the center of the black hole? (b) What is the mass of this black hole, assuming circular orbits? Express your answer in kilograms and as a multiple of our sun's mass. (c) What is the radius of its event horizon?

PROBLEMS

13.45 • Three uniform spheres are fixed at the positions shown in **Fig. P13.45**. (a) What are the magnitude and direction of the force on a 0.0150 kg particle placed at P ? (b) If the spheres are in deep outer space and a 0.0150 kg particle is released from rest 300 m from the origin along a line 45° below the $-x$ -axis, what will the particle's speed be when it reaches the origin?

Figure P13.45



13.46 ••• CP Exploring Europa. There is strong evidence that Europa, a satellite of Jupiter, has a liquid ocean beneath its icy surface. Many scientists think we should land a vehicle there to search for life. Before launching it, we would want to test such a lander under the gravity conditions at the surface of Europa. One way to do this is to put the lander at the end of a rotating arm in an orbiting earth satellite. If the arm is 4.25 m long and pivots about one end, at what angular speed (in rpm) should it spin so that the acceleration of the lander is the same as the acceleration due to gravity at the surface of Europa? The mass of Europa is $4.80 \times 10^{22} \text{ kg}$ and its diameter is 3120 km .

13.47 ••• CP An experiment is performed in deep space with two uniform spheres, one with mass 50.0 kg and the other with mass 100.0 kg . They have equal radii, $r = 0.20 \text{ m}$. The spheres are released from rest with their centers 40.0 m apart. They accelerate toward each other because of their mutual gravitational attraction. You can ignore all gravitational forces other than that between the two spheres. (a) Explain why linear momentum is conserved. (b) When their centers are 20.0 m apart, find (i) the speed of each sphere and (ii) the magnitude of the relative velocity with which one sphere is approaching the other. (c) How far from the initial position of the center of the 50.0 kg sphere do the surfaces of the two spheres collide?

13.48 •• Mission to Titan. On December 25, 2004, the *Huygens* probe separated from the *Cassini* spacecraft orbiting Saturn and began a 22 day journey to Saturn's moon Titan, on whose surface it landed. Besides the data in Appendix F, it is useful to know that Titan is $1.22 \times 10^6 \text{ km}$ from the center of Saturn and has a mass of $1.35 \times 10^{23} \text{ kg}$ and a diameter of 5150 km . At what distance from Titan should the gravitational pull of Titan the gravitational pull of Saturn?

13.49 • Geosynchronous Satellites. Many satellites are moving in a circle in the earth's equatorial plane. They are at such a height above the earth's surface that they always remain above the same point. (a) Find the altitude of these satellites above the earth's surface. (Such an orbit is said to be *geosynchronous*.) (b) Explain, with a sketch, why the radio signals from these satellites cannot directly reach receivers on earth that are north of 81.3° N latitude.

13.50 •• Two spherically symmetric planets with no atmosphere have the same average density, but planet B has twice the radius of planet A . A small satellite of mass m_A has period T_A when it orbits planet A in a circular orbit that is just above the surface of the planet. A small satellite of mass m_B has period T_B when it orbits planet B in a circular orbit that is just above the surface of the planet. How does T_B compare to T_A ?

13.51 ••• What is the escape speed from a 300-km-diameter asteroid with a density of 2500 kg/m^3 ?

13.52 ••• A landing craft with mass $12,500 \text{ kg}$ is in a circular orbit $5.75 \times 10^5 \text{ m}$ above the surface of a planet. The period of the orbit is 5800 s . The astronauts in the lander measure the diameter of the planet to be $9.60 \times 10^6 \text{ m}$. The lander sets down at the north pole of the planet. What is the weight of an 85.6 kg astronaut as he steps out onto the planet's surface?

13.53 •• Planet X rotates in the same manner as the earth, around an axis through its north and south poles, and is perfectly spherical. An astronaut who weighs 943.0 N on the earth weighs 915.0 N at the north pole of Planet X and only 850.0 N at its equator. The distance from the north pole to the equator is $18,850 \text{ km}$, measured along the surface of Planet X . (a) How long is the day on Planet X ? (b) If a $45,000 \text{ kg}$ satellite is placed in a circular orbit 2000 km above the surface of Planet X , what will be its orbital period?

13.54 ••• CALC Example 13.10 assumes a constant density for the earth, but Fig. 13.9 shows that this is not accurate. (a) Approximate the density graph in Fig. 13.9 by a straight line running from $13,000 \text{ kg/m}^3$ at $r = 0$ to 3000 kg/m^3 at $r = R_E$. Divide the earth into concentric shells of width dr and therefore volume $dV = 4\pi r^2 dr$. Integrate to find the mass enclosed as a function of the distance from the earth's center. (b) Find the expression for the magnitude of the gravitational force on a mass m at a distance r from the earth's center.

13.55 •• CP An astronaut, whose mission is to go where no one has gone before, lands on a spherical planet in a distant galaxy. As she stands on the surface of the planet, she releases a small rock from rest and finds that it takes the rock 0.480 s to fall 1.90 m . If the radius of the planet is $8.60 \times 10^7 \text{ m}$, what is the mass of the planet?

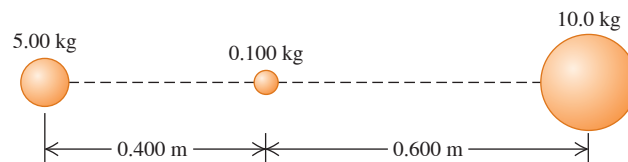
13.56 ••• CP Your starship, the *Aimless Wanderer*, lands on the mysterious planet Mongo. As chief scientist-engineer, you make the following measurements: A 2.50 kg stone thrown upward from the ground at 12.0 m/s returns to the ground in 4.80 s ; the circumference of Mongo at the equator is $2.00 \times 10^5 \text{ km}$; and there is no appreciable atmosphere on Mongo. The starship commander, Captain Confusion, asks for the following information: (a) What is the mass of Mongo? (b) If the *Aimless Wanderer* goes into a circular orbit $30,000 \text{ km}$ above the surface of Mongo, how many hours will it take the ship to complete one orbit?

13.57 •• CP You are exploring a distant planet. When your spaceship is in a circular orbit at a distance of 630 km above the planet's surface, the ship's orbital speed is 4900 m/s . By observing the planet, you determine its radius to be $4.48 \times 10^6 \text{ m}$. You then land on the surface and, at a place where the ground is level, launch a small projectile with initial speed 12.6 m/s at an angle of 30.8° above the horizontal. If resistance due to the planet's atmosphere is negligible, what is the horizontal range of the projectile?

13.58 •• The 0.100 kg sphere in Fig. P13.58 is released from rest at the position shown in the sketch, with its center 0.400 m from the center of the 5.00 kg mass. Assume that the only forces on the 0.100 kg sphere are the gravitational forces exerted by the other two spheres and that the

5.00 kg and 10.0 kg spheres are held in place at their initial positions. What is the speed of the 0.100 kg sphere when it has moved 0.400 m to the right from its initial position?

Figure P13.58



13.59 • The magnitude of the gravitational flux Φ_g through a surface with area A is defined such that it is equal to gA when the gravitational field $\vec{g}$ (see Exercise 13.40) is at all points on the surface constant and perpendicular to the surface. For a spherically symmetric object with radius R and mass M , this is the case for a surface that is spherical with radius r and concentric with the object. Consider the case where $r > R$. For this surface calculate the magnitude of the gravitational flux through the surface due to the gravitation field of the object. Does your answer depend on the radius r of the spherical surface? (We'll consider a similar result for the electric field in Chapter 22.)

13.60 ••• A narrow uniform rod has length $2a$. The linear mass density of the rod is ρ , so the mass m of a length l of the rod is ρl . (a) A point mass is located a perpendicular distance r from the center of the rod. Calculate the magnitude and direction of the force that the rod exerts on the point mass. (Hint: Let the rod be along the y -axis with the center of the rod at the origin, and divide the rod into infinitesimal segments that have length dy and that are located at coordinate y . The mass of the segment is $dm = \rho dy$. Write expressions for the x - and y -components of the force on the point mass, and integrate from $-a$ to $+a$ to find the components of the total force. Use the integrals in Appendix B.) (b) What does your result become for $a \gg r$? (Hint: Use the power series for $(1 + x)^n$ given in Appendix B.) (c) For $a \gg r$, what is the gravitational field $\vec{g} = \vec{F}_g/m$ at a distance r from the rod? (d) Consider a cylinder of radius r and length L whose axis is along the rod. As in part (c), let the length of the rod be much greater than both the radius and length of the cylinder. Then the gravitational field is constant on the curved side of the cylinder and perpendicular to it, so the gravitational flux Φ_g through this surface is equal to gA , where $A = 2\pi rL$ is the area of the curved side of the cylinder (see Problem 13.59). Calculate this flux. Write your result in terms of the mass M of the portion of the rod that is inside the cylindrical surface. How does your result depend on the radius of the cylindrical surface?

13.61 • Falling Hammer. A hammer with mass m is dropped from rest from a height h above the earth's surface. This height is not necessarily small compared with the radius R_E of the earth. Ignoring air resistance, derive an expression for the speed v of the hammer when it reaches the earth's surface. Your expression should involve h , R_E , and m_E (the earth's mass).

13.62 • (a) Calculate how much work is required to launch a spacecraft of mass m from the surface of the earth (mass m_E , radius R_E) and place it in a circular *low earth orbit*—that is, an orbit whose altitude above the earth's surface is much less than R_E . (As an example, the International Space Station is in low earth orbit at an altitude of about 400 km , much less than $R_E = 6370 \text{ km}$.) Ignore the kinetic energy that the spacecraft has on the ground due to the earth's rotation. (b) Calculate the minimum amount of additional work required to move the spacecraft from low earth orbit to a very great distance from the earth. Ignore the gravitational effects of the sun, the moon, and the other planets. (c) Justify the statement "In terms of energy, low earth orbit is halfway to the edge of the universe."

13.63 • Binary Star—Equal Masses. Two identical stars with mass M orbit around their center of mass. Each orbit is circular and has radius R , so that the two stars are always on opposite sides of the circle. (a) Find the gravitational force of one star on the other. (b) Find the orbital speed of each star and the period of the orbit. (c) How much energy would be required to separate the two stars to infinity?

13.64 • CP Binary Star—Different Masses. Two stars, with masses M_1 and M_2 , are in circular orbits around their center of mass. The star with mass M_1 has an orbit of radius R_1 ; the star with mass M_2 has an orbit of radius R_2 . (a) Show that the ratio of the orbital radii of the two stars equals the reciprocal of the ratio of their masses—that is, $R_1/R_2 = M_2/M_1$. (b) Explain why the two stars have the same orbital period, and show that the period T is given by $T = 2\pi(R_1 + R_2)^{3/2}/\sqrt{G(M_1 + M_2)}$. (c) The two stars in a certain binary star system move in circular orbits. The first star, Alpha, has an orbital speed of 36.0 km/s. The second star, Beta, has an orbital speed of 12.0 km/s. The orbital period is 137 d. What are the masses of each of the two stars? (d) One of the best candidates for a black hole is found in the binary system called A0620-0090. The two objects in the binary system are an orange star, V616 Monocerotis, and a compact object believed to be a black hole (see Fig. 13.28). The orbital period of A0620-0090 is 7.75 hours, the mass of V616 Monocerotis is estimated to be 0.67 times the mass of the sun, and the mass of the black hole is estimated to be 3.8 times the mass of the sun. Assuming that the orbits are circular, find the radius of each object's orbit and the orbital speed of each object. Compare these answers to the orbital radius and orbital speed of the earth in its orbit around the sun.

13.65 •• Comets travel around the sun in elliptical orbits with large eccentricities. If a comet has speed 2.0×10^4 m/s when at a distance of 2.5×10^{11} m from the center of the sun, what is its speed when at a distance of 5.0×10^{10} m?

13.66 • The planet Uranus has a radius of 25,360 km and a surface acceleration due to gravity of 9.0 m/s^2 at its poles. Its moon Miranda (discovered by Kuiper in 1948) is in a circular orbit about Uranus at an altitude of 104,000 km above the planet's surface. Miranda has a mass of 6.6×10^{19} kg and a radius of 236 km. (a) Calculate the mass of Uranus from the given data. (b) Calculate the magnitude of Miranda's acceleration due to its orbital motion about Uranus. (c) Calculate the acceleration due to Miranda's gravity at the surface of Miranda. (d) Do the answers to parts (b) and (c) mean that an object released 1 m above Miranda's surface on the side toward Uranus will fall up relative to Miranda? Explain.

13.67 •• CP Consider a spacecraft in an elliptical orbit around the earth. At the low point, or perigee, of its orbit, it is 400 km above the earth's surface; at the high point, or apogee, it is 4000 km above the earth's surface. (a) What is the period of the spacecraft's orbit? (b) Using conservation of angular momentum, find the ratio of the spacecraft's speed at perigee to its speed at apogee. (c) Using conservation of energy, find the speed at perigee and the speed at apogee. (d) It is necessary to have the spacecraft escape from the earth completely. If the spacecraft's rockets are fired at perigee, by how much would the speed have to be increased to achieve this? What if the rockets were fired at apogee? Which point in the orbit is more efficient to use?

13.68 •• A rocket with mass 5.00×10^3 kg is in a circular orbit of radius 7.20×10^6 m around the earth. The rocket's engines fire for a period of time to increase that radius to 8.80×10^6 m, with the orbit again circular. (a) What is the change in the rocket's kinetic energy? Does the kinetic energy increase or decrease? (b) What is the change in the rocket's gravitational potential energy? Does the potential energy increase or decrease? (c) How much work is done by the rocket engines in changing the orbital radius?

13.69 ••• A 5000 kg spacecraft is in a circular orbit 2000 km above the surface of Mars. How much work must the spacecraft engines perform to move the spacecraft to a circular orbit that is 4000 km above the surface?

13.70 •• A satellite with mass 848 kg is in a circular orbit with an orbital speed of 9640 m/s around the earth. What is the new orbital speed after friction from the earth's upper atmosphere has done -7.50×10^9 J of work on the satellite? Does the speed increase or decrease?

13.71 ••• CALC Planets are not uniform inside. Normally, they are densest at the center and have decreasing density outward toward the surface. Model a spherically symmetric planet, with the same radius as the earth, as having a density that decreases linearly with distance from the center. Let the density be $15.0 \times 10^3 \text{ kg/m}^3$ at the center and $2.0 \times 10^3 \text{ kg/m}^3$ at the surface. What is the acceleration due to gravity at the surface of this planet?

13.72 •• One of the brightest comets of the 20th century was Comet Hyakutake, which passed close to the sun in early 1996. The orbital period of this comet is estimated to be about 30,000 years. Find the semi-major axis of this comet's orbit. Compare it to the average sun–Pluto distance and to the distance to Alpha Centauri, the nearest star to the sun, which is 4.3 light-years distant.

13.73 ••• CALC An object in the shape of a thin ring has radius a and mass M . A uniform sphere with mass m and radius R is placed with its center at a distance x to the right of the center of the ring, along a line through the center of the ring, and perpendicular to its plane (see Fig. E13.39). What is the gravitational force that the sphere exerts on the ring-shaped object? Show that your result reduces to the expected result when x is much larger than a .

13.74 •• CALC A uniform wire with mass M and length L is bent into a semicircle. Find the magnitude and direction of the gravitational force this wire exerts on a point with mass m placed at the center of curvature of the semicircle.

13.75 • CALC A shaft is drilled from the surface to the center of the earth (see Fig. 13.25). As in Example 13.10 (Section 13.6), make the unrealistic assumption that the density of the earth is uniform. With this approximation, the gravitational force on an object with mass m , that is inside the earth at a distance r from the center, has magnitude $F_g = GmEmr/R_E^3$ (as shown in Example 13.10) and points toward the center of the earth. (a) Derive an expression for the gravitational potential energy $U(r)$ of the object–earth system as a function of the object's distance from the center of the earth. Take the potential energy to be zero when the object is at the center of the earth. (b) If an object is released in the shaft at the earth's surface, what speed will it have when it reaches the center of the earth?

13.76 •• DATA For each of the eight planets Mercury to Neptune, the semi-major axis a of their orbit and their orbital period T are as follows:

Planet	Semi-major Axis (10^6 km)	Orbital Period (days)
Mercury	57.9	88.0
Venus	108.2	224.7
Earth	149.6	365.2
Mars	227.9	687.0
Jupiter	778.3	4331
Saturn	1426.7	10,747
Uranus	2870.7	30,589
Neptune	4498.4	59,800

(a) Explain why these values, when plotted as T^2 versus a^3 , fall close to a straight line. Which of Kepler's laws is being tested? However, the values of T^2 and a^3 cover such a wide range that this plot is not a very practical way to graph the data. (Try it.) Instead, plot $\log(T)$ (with T in seconds) versus $\log(a)$ (with a in meters). Explain why the data should also fall close to a straight line in such a plot. (b) According to Kepler's laws, what should be the slope of your $\log(T)$ versus $\log(a)$ graph in part (a)? Does your graph have this slope? (c) Using $G = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$, calculate the mass of the sun from the y-intercept of your graph. How does your calculated value compare with the value given in Appendix F? (d) The only asteroid visible to the naked eye (and then only under ideal viewing conditions) is Vesta, which has an orbital period of 1325.4 days. What is the length of the semi-major axis of Vesta's orbit? Where does this place Vesta's orbit relative to the orbits of the eight major planets? Some scientists argue that Vesta should be called a minor planet rather than an asteroid.

13.77 •• DATA For a spherical planet with mass M , volume V , and radius R , derive an expression for the acceleration due to gravity at the planet's surface, g , in terms of the average density of the planet, $\rho = M/V$, and the planet's diameter, $D = 2R$. The table gives the values of D and g for the eight major planets:

Planet	D (km)	g (m/s ²)
Mercury	4879	3.7
Venus	12,104	8.9
Earth	12,756	9.8
Mars	6792	3.7
Jupiter	142,984	23.1
Saturn	120,536	9.0
Uranus	51,118	8.7
Neptune	49,528	11.0

(a) Treat the planets as spheres. Your equation for g as a function of ρ and D shows that if the average density of the planets is constant, a graph of g versus D will be well represented by a straight line. Graph g as a function of D for the eight major planets. What does the graph tell you about the variation in average density? (b) Calculate the average density for each major planet. List the planets in order of decreasing density, and give the calculated average density of each. (c) The earth is not a uniform sphere and has greater density near its center. It is reasonable to assume this might be true for the other planets. Discuss the effect this has on your analysis. (d) If Saturn had the same average density as the earth, what would be the value of g at Saturn's surface?

13.78 ••• DATA For a planet in our solar system, assume that the axis of orbit is at the sun and is circular. Then the angular momentum about that axis due to the planet's orbital motion is $L = MvR$. (a) Derive an expression for L in terms of the planet's mass M , orbital radius R , and period T of the orbit. (b) Using Appendix F, calculate the magnitude of the orbital angular momentum for each of the eight major planets. (Assume a circular orbit.) Add these values to obtain the total angular momentum of the major planets due to their orbital motion. (All the major planets orbit in the same direction in close to the same plane, so adding the magnitudes to get the total is a reasonable approximation.) (c) The rotational period of the sun is 24.6 days. Using Appendix F, calculate the angular momentum the sun has due to the rotation about its axis. (Assume that the sun is a uniform sphere.) (d) How does the rotational angular momentum of the sun compare with the total orbital angular momentum of the planets? How does the mass of the sun compare with the total mass of the planets? The fact that the sun has most of

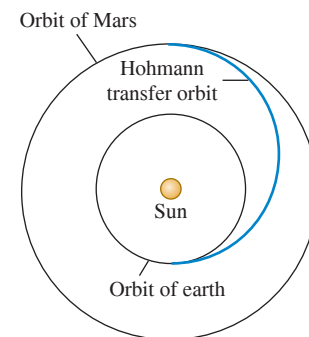
the mass of the solar system but only a small fraction of its total angular momentum must be accounted for in models of how the solar system formed. (e) The sun has a density that decreases with distance from its center. Does this mean that your calculation in part (c) overestimates or underestimates the rotational angular momentum of the sun? Or doesn't the nonuniform density have any effect?

CHALLENGE PROBLEMS

13.79 ••• Interplanetary

Navigation. The most efficient way to send a spacecraft from the earth to another planet is to use a *Hohmann transfer orbit* (Fig. P13.79). If the orbits of the departure and destination planets are circular, the Hohmann transfer orbit is an elliptical orbit whose perihelion and aphelion are tangent to the orbits of the two planets. The rockets are fired briefly at the departure planet to put the spacecraft into the transfer orbit; the spacecraft then coasts until it reaches the destination planet. The rockets are then fired again to put the spacecraft into the same orbit about the sun as the destination planet. (a) For a flight from earth to Mars, in what direction must the rockets be fired at the earth and at Mars: in the direction of motion or opposite the direction of motion? What about for a flight from Mars to the earth? (b) How long does a one-way trip from the earth to Mars take, between the firings of the rockets? (c) To reach Mars from the earth, the launch must be timed so that Mars will be at the right spot when the spacecraft reaches Mars's orbit around the sun. At launch, what must the angle between a sun–Mars line and a sun–earth line be? Use Appendix F.

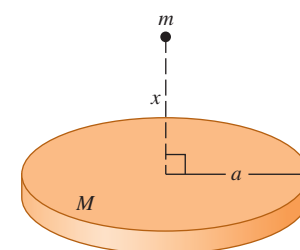
Figure P13.79



13.80 ••• CP Tidal Forces near a Black Hole. An astronaut inside a spacecraft, which protects her from harmful radiation, is orbiting a black hole at a distance of 120 km from its center. The black hole is 5.00 times the mass of the sun and has a Schwarzschild radius of 15.0 km. The astronaut is positioned inside the spaceship such that one of her 0.030 kg ears is 6.0 cm farther from the black hole than the center of mass of the spacecraft and the other ear is 6.0 cm closer. (a) What is the tension between her ears? Would the astronaut find it difficult to keep from being torn apart by the gravitational forces? (Since her whole body orbits with the same angular velocity, one ear is moving too slowly for the radius of its orbit and the other is moving too fast. Hence her head must exert forces on her ears to keep them in their orbits.) (b) Is the center of gravity of her head at the same point as the center of mass? Explain.

13.81 ••• CALC Mass M is distributed uniformly over a disk of radius a . Find the gravitational force (magnitude and direction) between this disk-shaped mass and a particle with mass m located a distance x above the center of the disk (Fig. P13.81). Does your result reduce to the correct expression as x becomes very large? (Hint: Divide the disk into infinitesimally thin concentric rings, use the expression derived in Exercise 13.39 for the gravitational force due to each ring, and integrate to find the total force.)

Figure P13.81



MCAT-STYLE PASSAGE PROBLEMS

Exoplanets. As planets with a wide variety of properties are being discovered outside our solar system, astrobiologists are considering whether and how life could evolve on planets that might be very different from earth. One recently discovered extrasolar planet, or exoplanet, orbits a star whose mass is 0.70 times the mass of our sun. This planet has been found to have 2.3 times the earth's diameter and 7.9 times the earth's mass. For planets in this size range, computer models indicate a relationship between the planet's density and composition:

Density Compared with That of the Earth	Composition
2–3 times	Mostly iron
0.9–2 times	Iron core with a rock mantle
0.4–0.9 times	Iron core with a rock mantle and some lighter elements, such as (water) ice
<0.4 times	Hydrogen and/or helium gas

Based on S. Seager et al., "Mass–Radius Relationships for Solid Exoplanets"; arXiv:0707.2895 [astro-ph].

13.82 Based on these data, what is the most likely composition of this planet? (a) Mostly iron; (b) iron and rock; (c) iron and rock with some lighter elements; (d) hydrogen and helium gases.

13.83 How many times the acceleration due to gravity g near the earth's surface is the acceleration due to gravity near the surface of this exoplanet? (a) About 0.29 g ; (b) about 0.65 g ; (c) about 1.5 g ; (d) about 7.9 g .

13.84 Observations of this planet over time show that it is in a nearly circular orbit around its star and completes one orbit in only 9.5 days. How many times the orbital radius r of the earth around our sun is this exoplanet's orbital radius around its sun? Assume that the earth is also in a nearly circular orbit. (a) 0.026 r ; (b) 0.078 r ; (c) 0.70 r ; (d) 2.3 r .

ANSWERS

Chapter Opening Question ?

(iv) For a satellite a distance r from the center of its planet, the orbital speed is proportional to $\sqrt{1/r}$ and the acceleration due to gravity is proportional to $1/r^2$ (see Section 13.4). Hence a particle that orbits close to Saturn has a faster speed and a greater acceleration than one that orbits farther away.

Key Example VARIATION Problems

VP13.5.1 2.99×10^{10} J

VP13.5.2 6.46×10^3 m/s

VP13.5.3 1.88×10^6 m

VP13.5.4 1.39×10^4 m/s

VP13.6.1 (a) 2.49×10^7 m (b) 1.85×10^7 m (c) 10.9 h

VP13.6.2 (a) 2.04×10^7 m (b) 1.45×10^3 m/s

VP13.6.3 (a) $+1.72 \times 10^{11}$ J (b) -3.44×10^{11} J
(c) -1.72×10^{11} J

VP13.6.4 (a) -1.19×10^{10} J (b) $+1.83 \times 10^{10}$ J

VP13.9.1 (a) 1.50×10^{12} m (b) 0.996 (c) 31.8 y

VP13.9.2 (a) 5.91×10^{11} m (b) 4.43×10^{11} m (c) 7.39×10^{11} m

VP13.9.3 (a) 5.9×10^{12} m (b) 4.4×10^{12} m; inside

VP13.9.4 (a) 6.13×10^9 m (b) 1.67×10^{30} kg

Bridging Problem

(a) Perihelion: $v_P = \sqrt{\frac{Gm_S}{a} \frac{(1+e)}{(1-e)}}$

aphelion: $v_A = \sqrt{\frac{Gm_S}{a} \frac{(1-e)}{(1+e)}}$

(b) $v_P = 54.4$ km/s, $v_A = 0.913$ km/s; $K_P = 3.26 \times 10^{23}$ J,
 $U_P = -3.31 \times 10^{23}$ J, $E_P = -5.47 \times 10^{21}$ J;
 $K_A = 9.17 \times 10^{19}$ J, $U_A = -5.56 \times 10^{21}$ J,
 $E_A = -5.47 \times 10^{21}$ J